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Accelerating Trust-Region Methods: An Attempt to Balance Global and Local Efficiency

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Abstract. Balancing global efficiency and local convergence remains a central challenge in second-order optimization for unconstrained convex optimization problems. Newton's method enjoys fast local convergence but may diverge when initialized far from the solution [18, 44]. In contrast, accelerated second-order methods provide global guarantees but typically suffer from slower local convergence [7, 12, 30]. This raises the fundamental question of to what extent global acceleration can be achieved without sacrificing strong local convergence. In this paper, we tackle this challenge by proposing the first accelerated trust-region-type methods and leveraging their inherent primal-dual information. Our primary contribution is *Accelerating with Local Detection*, which utilizes the Lagrange multiplier to detect local regions and achieves a global oracle complexity of $\tilde{O}(\epsilon^{-1/3})$, while maintaining quadratic local convergence. We further examine the trade-off that arises when global convergence is pushed to the limit. Specifically, we introduce the *Accelerated Trust-Region Extragradient Method*, which achieves a global oracle complexity of $\tilde{O}(\epsilon^{-2/7})$ but no longer enjoys quadratic local convergence. This establishes a phase transition in accelerated trust-region-type methods: quadratic local convergence is preserved under moderate global acceleration, but it breaks down when pursuing extreme global efficiency. Numerical experiments further confirm the theoretical results.

Key words: Trust-Region Method, Acceleration, Global Oracle Complexity, Local Analysis

1. Introduction In this paper, we consider the unconstrained convex optimization problem

$$f^* = \min_{x \in \mathbb{R}^n} f(x), \quad \text{eq.main problem (1.1)}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex function. When the Hessian of the objective is available, second-order methods (SOMs) are particularly effective for solving (1.1), especially when high-accuracy solutions are desired. Their power stems from a central property: Newton's method achieves quadratic convergence in a neighborhood of any non-degenerate local minimum [44, 50, 53]. Many SOMs are thus designed to approximate the Newton direction, allowing them to inherit its excellent local convergence behavior.

In addition to local convergence, another important criterion for evaluating the effectiveness of optimization methods is their non-asymptotic global oracle complexity, which refers to the number of second-order subproblems (the oracle calls) needed to find an approximate solution [11]. Over the past decades, a few SOMs have been developed with such emphases [9, 10, 15, 16, 25, 26, 35, 41, 49]. In convex optimization, some of these methods can be further improved through acceleration techniques.

In this context, two major approaches have emerged. The first approach, accelerated cubic regularized Newton (CRN) method [46], achieves a global oracle complexity of $O(\epsilon^{-1/3})$ based on the cubic regularization oracle [49]. A different strategy is adopted in the accelerated Newton proximal extragradient (A-NPE) method [42], which improves the oracle complexity to $\tilde{O}(\epsilon^{-2/7})$ and is compatible with second-order oracles that satisfy certain error bound conditions [7]. These developments, largely based on the estimating sequence technique [5, 45], have inspired a wide range of enhancements [3, 4, 12, 19, 29, 31, 34, 39]. Recently, the oracle complexity of the A-NPE method was improved to be an optimal rate in $O(\epsilon^{-2/7})$ [7, 36]. These acceleration techniques are broadly applicable and are by no means limited to the domain of SOMs [21, 37, 62, 63].

However, it is hard to balance the global and local efficiency of second-order optimization algorithms. For instance, the classical Newton's method possesses excellent local convergence but lacks global guarantees, often exhibiting divergence when the starting point is far from the optimal solution [18, 44]. In contrast, stronger worst-case complexity results of accelerated SOMs often come at the cost of local efficiency. Empirical evidence shows that in many tasks like logistic regression, accelerated SOMs are often outperformed by unaccelerated ones [7, 12, 29, 30], and manually switching to Newton's method is often necessary to obtain highly accurate solutions [12, 29, 30]. The above phenomenon underscores a fundamental dilemma in accelerated SOMs, and raises the following question:

How much can we accelerate SOMs globally while maintaining their excellent local convergence?

1.1. Motivation and our approach sec.motivation It is folklore that the strong local performance of SOMs ultimately stems from the Newton step. Thus, to design an accelerated method with excellent local performance, the second-order oracle should have the ability to detect the local geometry near the optimal solution, providing a signal indicating that the iterates may have entered the quadratic convergence region, allowing a Newton step to be active. For that sake, one particular choice could be the trust-region (TR) oracle [13, 43]. At a given point $x \in \mathbb{R}^n$, the classic TR oracle solves the following subproblem:

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \quad & \nabla f(x)^T d + \frac{1}{2} d^T \nabla^2 f(x) d \\ \text{s.t.} \quad & \|d\| \leq r, \end{aligned} \quad \text{eq.classic tr subproblem (1.2)}$$

where r denotes the trust-region radius. We denote $(d, \lambda) = \text{TR}(x, r)$ as the primal-dual solution to this problem. For an overview of TR methods, we refer readers to two excellent monographs [67, 68].

Two features make (1.2) particularly appealing. Near a non-degenerate minimizer, the step reduces to the pure Newton direction once the ball constraint becomes inactive, preserving quadratic convergence, which is the capability absent from other second-order oracles. Moreover, TR oracles can be solved in $O(n^3 + n^2 \log \log(\epsilon^{-1}))$ arithmetic operations [60, 65], incurring no significant overhead compared to solving a Newton equation. On the practical side, the TR oracle has been thoroughly analyzed [27, 32, 55, 57, 58, 61], so that highly competitive solvers have been developed [1, 2, 55] for it and its extensions. To this end, TR oracles have formed the algorithmic core of general-purpose nonlinear programming solvers such as Knitro [6], COPT [20], and PDFO [54]. Similar success can be found in other cross-cutting applications ranging from adversarial training [64] to reinforcement learning [56]. These advantages serve as the motivation to use TR to address the global-local balance in accelerated SOMs.

However, to our best knowledge, TR methods have not been accelerated yet as using TR oracles introduces technical challenges. Classical TR methods cannot match the global oracle complexity of other unaccelerated SOMs [8, 16] because the dual variable λ associated with the ball constraint in (1.2) is determined a posteriori and requires careful monitoring [14]. Achieving global acceleration demands relative stability between this multiplier and the step size, which roughly means that λ should not vary too fast relative to the step size, but enforcing such stability can interfere with transitions to Newton steps near the optimum.

To address this challenge, we design, for the first time, the accelerated trust-region methods that leverage a modified TR oracle, which we call the trust-region oracle for acceleration denoted as TR_+ . In particular, at a point $x \in \mathbb{R}^n$, the TR_+ oracle solves

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \quad & \nabla f(x)^T d + \frac{1}{2} d^T (\nabla^2 f(x) + \sigma I) d, \\ \text{s.t.} \quad & \|d\| \leq r. \end{aligned} \quad (\text{TR}_+)$$

The dual variable associated with the ball constraint is denoted by λ , which can detect local geometric information of the optimal solution. The primal regularization σ in TR_+ is used to effectively modulate λ , and maintains the stability required for global acceleration, enabling our accelerated trust-region method to systematically balance fast local convergence with provable global guarantee.

1.2. Contribution This work provides an answer to the question posed earlier: we demonstrate that the compatibility between the global and local efficiency of the SOMs depends on the degree of global acceleration. Specifically, based on the TR_+ oracles [35], we develop two accelerated TR-type methods for problem (1.1). One can simultaneously achieve both global acceleration and quadratic local convergence, and the other has a faster global convergence rate but loses local efficiency.

The first method, *Accelerated Trust-Region Method with Local Detection* (Alg. 1), implements a local detection mechanism (Subroutine 1) that automatically identifies and “dives into” the quadratic convergence regions. This design yields an improved global worst-case oracle complexity of $\tilde{O}(\epsilon^{-1/3})$, while achieving a local quadratic rate of convergence. To our best knowledge, this is the first accelerated SOM that provably achieves both $\tilde{O}(\epsilon^{-1/3})$ global oracle complexity and local quadratic convergence.

The second method, *Accelerated Trust-Region Extragradient Method* (Alg. 2), is designed to pursue the limits of global performance, reaching the near-optimal global oracle complexity of $\tilde{O}(\epsilon^{-2/7})$, but loses the excellent local convergence. Thus, our analysis and experiments reveal a phase transition in accelerated trust-region type methods: pushing global efficiency to its limits naturally entails a trade-off with local performance. Such a phase transition occurs due to the loss of local detection in Alg. 2 when pursuing the near-optimal global convergence rate, where the primal regularizer σ is set prior to determining the extrapolation point that is used in the TR_+ oracle (as in Alg. 1).

Our theoretical findings are corroborated by numerical experiments. On a global scale, all accelerated SOMs, including ours and [42, 46], consistently outperform the non-accelerated trust-region method [35] and cubic Newton method [49]. Locally, Alg. 1 exhibits quadratic convergence, and Alg. 2 cannot retain superlinear convergence near the solution, similar to the empirical conclusions in [7, 29, 34]. These findings again confirm the necessity of local detection.

1.3. Related works We review related works along two main lines: TR-type methods and accelerated SOMs that aim to achieve faster-than-sublinear efficiency. Despite the strong empirical success of TR-type methods, their theoretical analysis mainly focuses on nonconvex problems [15–17, 22–24, 52, 66, 69], while their global oracle complexity analysis for convex problems remains incomplete. In fact, in convex optimization, the classical TR has worse convergence rates than other unaccelerated mainstream SOMs. This gap was recently closed in [35]. However, whether TR oracles can further benefit from acceleration, as achieved for other SOMs, remains an open question.

As discussed earlier, unaccelerated SOMs often exhibit superior local performance near the optimal solution compared with their accelerated counterparts. A common remedy for this issue is to employ restart strategies in acceleration frameworks [47]. However, this introduces a new challenge of deciding when to restart since the optimal restart frequency depends on unknown, problem-specific parameters. In contrast, when stronger global regularity conditions (akin to strong convexity) are imposed on the objective function, the restart schedule can be determined in a more structured form [28, 38, 51] or even becomes unnecessary [33, 40]. Under such assumptions, these methods can achieve strong global performance, and thus the intrinsic trade-off between global convergence and local efficiency becomes a secondary concern.

Organization of the paper The remaining part of this paper is organized as follows. Section 2 reviews the background on TR-type oracles and the use of estimate sequence techniques in accelerated second-order optimization. In Section 3, we present the Alg. 1, which achieves an oracle complexity of $\tilde{O}(\epsilon^{-1/3})$ and exhibits quadratic convergence in high-accuracy regimes. Section 4 introduces the Alg. 2, which explores the phase transition of acceleration and attains a near-optimal oracle complexity of $\tilde{O}(\epsilon^{-2/7})$. Section 5 reports numerical results on practical tasks, confirming the theoretical advantages of our proposed methods.

2. Preliminaries Throughout the paper, $|\cdot|$ denotes the standard Euclidean norm in $\mathbb{R}^n$. We use $x^T y$ and $\langle x, y \rangle$ interchangeably to denote the inner product between $x, y \in \mathbb{R}^n$. For a matrix $X \in \mathbb{R}^{n \times n}$, $|X|$ denotes the induced ℓ_2 norm. Unless otherwise stated, $\log(\cdot)$ denotes the base-2 logarithm, while $\ln(\cdot)$ denotes the natural logarithm. **In the statements of technical results, symbols such as x_k, v_k, y_k , etc., refer to the iterates and steps generated by the algorithm, rather than to generic vectors.**

We are interested in finding approximate solutions of (1.1) defined as below, which is also used in Monteiro and Svaiter [42].

DEFINITION 2.1. Let $0 < \epsilon_f < 1$ and $0 < \epsilon_g < 1$ be two prescribed tolerances. A point $x \in \mathbb{R}^n$ is called an ϵ_f -function-value solution of problem (1.1) if

$$f(x) - f^* \leq \epsilon_f. \quad \text{eq.function-value-sol (2.1)}$$

A point $x \in \mathbb{R}^n$ is called an ϵ_g -stationary-solution of problem (1.1) if

$$\|\nabla f(x)\| \leq \epsilon_g. \quad \text{eq.first-order-stat (2.2)}$$

A point $x \in \mathbb{R}^n$ is called an (ϵ_f, ϵ_g) -approximate solution of problem (1.1) if it satisfies (2.1) or (2.2).

The following assumptions are used throughout the paper. Some of them are invoked only when necessary and will be stated explicitly. **Note that Assumption 2.3 is equivalent to the gradient of the objective function being Lipschitz.**

ASSUMPTION 2.1. The objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice continuously differentiable, and its Hessian is Lipschitz continuous. That is, there exists a constant $M > 0$ such that for all $x, y \in \mathbb{R}^n$,

$$\|\nabla^2 f(x) - \nabla^2 f(y)\| \leq M\|x - y\|. \quad \text{eq.main assm (2.3)}$$

ASSUMPTION 2.2. The problem (1.1) is solvable; that is, there exists $x^* \in \mathbb{R}^n$ such that

$$f(x^*) = f^* := \min_{x \in \mathbb{R}^n} f(x).$$

ASSUMPTION 2.3. The Hessian of the objective function is bounded. That is, there exists a constant $\kappa_H > 0$ such that for all $x \in \mathbb{R}^n$,

$$\|\nabla^2 f(x)\| \leq \kappa_H. \quad \text{eq.bounded hessian (2.4)}$$

As a direct consequence of Assumption 2.1, we have

LEMMA 2.1 (Lemma 4.1.1, Nesterov [47]). If $f : \mathbb{R}^n \mapsto \mathbb{R}$ satisfies Assumption 2.1, then for all $x, y \in \mathbb{R}^n$, we have

$$\|\nabla f(y) - \nabla f(x) - \nabla^2 f(x)(y - x)\| \leq \frac{M}{2}\|y - x\|^2 \quad \text{eq.first-order exp (2.5a)}$$

$$\left| f(y) - f(x) - \nabla f(x)^T(y - x) - \frac{1}{2}(y - x)^T \nabla^2 f(x)(y - x) \right| \leq \frac{M}{6}\|y - x\|^3. \quad \text{eq.second-order exp (2.5b)}$$

To proceed, let us introduce a unified view of the updates of SOMs. In fact, almost all of the SOMs choose the step d at a given point x as

$$(\nabla^2 f(x) + \mu I)d = -\nabla f(x). \quad \text{eq.second-order update (2.6)}$$

The regularization parameter μ plays a critical role in the convergence analysis and varies among different SOMs. For example, in the classical TR method using the standard oracle (1.2), μ is fully determined as a posteriori, and is exactly the Lagrangian multiplier associated with the ball constraint in (1.2). In the regularized Newton methods [41], μ can be explicitly selected as $\Theta(\sqrt{\|\nabla f(x)\|})$ such that an $O(\epsilon_f^{-1/2})$ method for finding ϵ_f -function-value solutions can be designed. In the CRN method [49], it requires some search procedures on μ .

The following lemmas clarify μ plays an important role in the convergence analysis of SOMs, the proofs of which can be found in Appendix A.

LEMMA 2.2. Suppose Assumption 2.1 and (2.6) hold, then we have

$$\|\nabla f(x + d)\| \leq \frac{M}{2}\|d\|^2 + \mu\|d\|. \quad \text{eq.bound next gradient (2.7)}$$

LEMMA 2.3. Suppose Assumption 2.1 holds and that (2.6) hold for some $\mu \geq M\|d\|$, then we have

$$\langle \nabla f(x+d), -d \rangle \geq \frac{\|\nabla f(x+d)\|^2}{2\mu} + \frac{3}{8}\mu\|d\|^2. \quad \text{eq. inner product } g, d \quad (2.8)$$

Further, if $\mu \leq 2M\|d\|$, we have

$$\langle \nabla f(x+d), -d \rangle \geq \frac{\sqrt{6}}{6} \frac{\|\nabla f(x+d)\|^{3/2}}{\sqrt{M}}. \quad \text{eq. inner product } g, d \text{ advanced} \quad (2.9)$$

Generally speaking, Lemma 2.2 shows that we can bound the next-iterate gradient norm by controlling μ . Lemma 2.3 serves as a key bridge to establish the global efficiency (in fact, it is a modified version of [47, Lemma 4.2.5] and [48, Corollary 1] for TR-type oracle), which suggests that μ should be selected roughly as $\Theta(\|d\|)$. This condition generally fails for the classical TR oracle, as explained in Section 1.1 and also discussed in [14, 16]. For this reason, we adopt the TR_+ oracle:

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \quad & \nabla f(x)^T d + \frac{1}{2} d^T (\nabla^2 f(x) + \sigma I) d, \\ \text{s.t.} \quad & \|d\| \leq r, \end{aligned} \quad \text{eq. newTR} \quad (\text{TR}_+)$$

We denote $(d, \lambda) = \text{TR}_+(x, \sigma, r)$ as the primal-dual solution of the TR_+ subproblem, and in the following, all references to solution pairs (d, λ) will refer to this TR_+ solution unless otherwise stated. The global optimality conditions are as follows.

LEMMA 2.4 (Section 3, Conn et al. [13]). The direction d is the solution to TR_+ if and only if there exists a dual multiplier $\lambda \geq 0$ such that:

$$\begin{aligned} \|d\| &\leq r & \text{eq. optcond primal} & (2.10a) \\ \lambda (\|d\| - r) &= 0 & \text{eq. optcond compl slack} & (2.10b) \\ (\nabla^2 f(x) + \sigma I + \lambda I) d &= -\nabla f(x) & \text{eq. optcond firstorder} & (2.10c) \\ \nabla^2 f(x) + \sigma I + \lambda I &\succeq 0. & \text{eq. optcond secondorder} & (2.10d) \end{aligned}$$

In the TR_+ oracle, the regularization parameter decomposes as $\mu = \sigma + \lambda$, aggregating a *prior* regularizer from the primal problem and a *posterior* dual variable. As we will show in the following, this primal-dual combo of information has the potential to balance global and local behavior.

3. Variant I: Balancing the Global-Local Trade-off In this section, we propose the first accelerated TR-type method, which applies the primal-dual structure of TR_+ into the estimating sequence [46]. The global oracle complexity for finding ϵ_f -approximate-solution is improved to $\tilde{O}(\epsilon_f^{-1/3})$ as opposed to the recent $O(\epsilon_f^{-1/2})$ in convex optimization [35], while the local rate of convergence to the non-degenerate solution remains quadratic.

3.1. Algorithm design Alg. 1 and its local detection mechanism (Subroutine 1) are presented below. The overall regularization in TR_+ is a combination of primal and dual information: $\mu_k = \sigma_k + \lambda_k$. The primal part σ_k is chosen carefully to make the dual part more stable and still carry local information, while the dual part λ_k is used as an indicator to check whether the current iterate is close to the local optimum.

The case $\lambda_k > 0$ typically indicates that the current iterate is still far from a local optimum. In that regime, as shown later in Lemma 3.1, the overall regularization parameter μ_k becomes automatically proportional to the step size, which justifies the updates of the estimating sequence in the acceleration framework.

The case $\lambda_k = 0$ is a signal that the current extrapolation point y_k may have entered the local quadratic convergence region of Newton's method (see Lemma 3.12). This triggers the Local Detection procedure (LD, Subroutine 1), which consists of two separate tracks named local diving and R&B, respectively, probing the local geometry around y_k . We first perform a simple examination in case the current solution is already ϵ_g -stationary-solution, see line 3 of Alg. 1 and Corollary 3.2.

In local diving (Track 1), we run Newton's method starting from y_k , the resulting sequence is indexed by i and denoted by $\{z_i\}$. As discussed in Section 3.3, if y_k indeed lies within the quadratic convergence

Algorithm 1 Accelerated Trust-Region Method with Local Detection (Alg. 1)

```

1: input: initial point  $x_0 = v_0 \in \mathbb{R}^n$ ,  $s_0 = 0$ , stationarity tolerance  $\epsilon_g > 0$ 
2: for  $k = 0, 1, 2, \dots$  do
3:    $y_k = \frac{k}{k+3}x_k + \frac{3}{k+3}v_k$ 
4:    $(\sigma_k, r_k) = (\frac{\sqrt{2M}}{2}\|\nabla f(y_k)\|^{1/2}, \frac{1}{\sqrt{2M}}\|\nabla f(y_k)\|^{1/2})$  |line.set-params
5:    $(d_k, \lambda_k) = \text{TR}_+(y_k, \sigma_k, r_k)$ 
6:    $x_{k+1} = y_k + d_k$ 
7:   if  $\lambda_k = 0$  then |line.if
8:     # Enter Local Detection (Subroutine 1)
9:      $(x_{k+1}, \text{ET}) = \text{LD}(y_k, \frac{\sqrt{2M}}{2}\|\nabla f(y_k)\|^{1/2}, d_k, \epsilon_g)$  |line.early terminate main
10:    if ET then
11:      terminate and output  $x_{k+1}$ 
12:    end if
13:  end if
14:   $s_{k+1} = s_k + \frac{(k+1)(k+2)}{2}\nabla f(x_{k+1})$  |line.update-s
15:   $v_{k+1} = v_0 - \sqrt{\frac{1}{24M\|s_{k+1}\|}}s_{k+1}$  |line.update-v
16: end for

```

Subroutine 1 Local Detection (LD)

```

1: input:  $z_0 = y \in \mathbb{R}^n$ ,  $\mu_+$ ,  $d_+$ ,  $\epsilon_g$ ; |line.right bracket point
2: if  $\|d_+\| \leq \min\{\frac{\epsilon_g}{2\kappa_H}, \sqrt{\frac{\epsilon_g}{M}}\}$  then |line.et 2
3:   output  $y + d_+$ , ET=True |line.if coro 3.1
4: end if
5: # Track 1: Local Diving
6: for  $i = 1, \dots, \lceil \log \frac{\|\nabla f(y)\|}{\epsilon_g} \rceil$  do |line.track 1 number of iter
7:    $z_i = z_{i-1} - \nabla^2 f(z_{i-1})^{-1}\nabla f(z_{i-1})$ 
8:   if  $\|\nabla f(z_i)\| \leq \epsilon_g$  then |line.et 1
9:     output  $z_i$ , ET=True
10:  end if
11: end for
12: # Track 2: Ratio Bracketing and Bisection (R&B)
13: if  $\frac{\mu_+}{\|d_+\|} \leq 2M$  then |line.check right bracket point
14:   output  $y + d_+$ , ET=False |# Check if the bisection is needed
15: else
16:    $r_- = \|d_+\|$ ,  $r_+ = \|(\nabla^2 f(y) + M\|d_+\|I)^{-1}\nabla f(y)\|$  |line.left bracketing point
17: end if
18: while  $\frac{\mu}{\|d\|} < M$  or  $\frac{\mu}{\|d\|} > 2M$  do
19:    $r = \frac{r_- + r_+}{2}$  |line.linear system oracle |# Perform bisection over  $r \in [r_-, r_+]$ 
20:    $(d, \mu) = \text{TR}_+(y, 0, r)$ 
21:   if  $\frac{\mu}{\|d\|} < M$  then
22:      $r_+ = r$ 
23:   else if  $\frac{\mu}{\|d\|} > 2M$  then
24:      $r_- = r$ 
25:   end if
26: end while
27: output  $y + d$ , ET=False

```

region, the resulting sequence $\{z_i\}$ will exhibit quadratic convergence, which justifies early termination of Alg. 1. (We could stop diving if the $\|\nabla f(z_i)\|$ is increasing, or a degenerate Hessian is detected.)

In R&B (Track 2), we attempt to find a step d_k whose size is proportional to the regularization parameter μ_k . This step certifies the updates of the estimate sequence and thus preserves global acceleration.

In summary, Alg. 1 admits two possible termination routes: when y_k enters the region of local quadratic convergence, the algorithm detects this and terminates early with ET=True (see lines 3 and 9 in Subroutine 1). Otherwise, it continues to follow a path of globally accelerated sequences with ET = False.

3.2. Global convergence in function value Now we analyze the worst-case global oracle complexity of Alg. 1 for computing an (ϵ_f, ϵ_g) -approximate solution as defined in Definition 2.1. Since our goal here is to analyze the worst-case global oracle complexity, we assume the iterates do not step into local diving of Subroutine 1, which is more relevant to local convergence analysis (cf. Section 3.3). Our analysis follows a standard procedure: we first establish the *iteration complexity* of the accelerated sequence in Alg. 1, and subsequently determine the number of oracle calls required per iteration in Subroutine 1, thereby obtaining the final *oracle complexity* bound.

3.2.1. Iterations complexity of Alg. 1 In this section, we show that Alg. 1 takes $O(\epsilon_f^{-1/3})$ to find an ϵ_f -function-value solution as in Definition 2.1.

When using the estimating sequence technique in [46], the globally accelerated convergence is guaranteed by maintaining the following two relations across iterations | eq.es relation

$$\phi_k^* \geq A_k f(x_k), \quad \phi_k^* = \phi_k(v_k) = \min_{x \in \mathbb{R}^n} \phi_k(x) \quad \text{eq.es relation 1} \quad (3.1a)$$

$$\phi_k(x) \leq A_k f(x) + \phi_0(x), \quad \forall x \in \mathbb{R}^n. \quad \text{eq.es relation 2} \quad (3.1b)$$

Here $\{\phi_k(x)\}_{k \geq 0}$ is a sequence of functions that approximate $f(x)$ from with $\{v_k\}_{k \geq 0}$ being the optimum, $\{A_k\}_{k \geq 0}$ is a sequence that measures the convergence rate of the sequence $\{x_k\}_{k \geq 0}$. As a result of (3.1), we have

$$f(x_k) - f(x) \leq \frac{1}{A_k} \phi_0(x), \quad \forall x \in \mathbb{R}^n, \quad k \geq 1.$$

Therefore, the global iteration complexity directly follows from the choice of A_k and $\phi_0(x)$.

For the upcoming analysis, we first complete the definition of $A_k, \phi_k(x)$ in the estimating sequence framework that was not explicitly presented in the algorithm:

$$a_k = \frac{(k+1)(k+2)}{2}, \quad A_k = \frac{k(k+1)(k+2)}{6}, \quad \text{eq.update a and A} \quad (3.2)$$

$$\phi_0(x) = 8M\|x - x_0\|^3, \quad \phi_{k+1}(x) = \phi_k(x) + a_k (f(x_{k+1}) + \langle \nabla f(x_{k+1}), x - x_{k+1} \rangle). \quad \text{eq.update phi} \quad (3.3)$$

We introduce some basic properties of the estimating sequence in Section A, which are from Nesterov [47]. To proceed, we categorize the iterates generated by Alg. 1 into two disjoint sets based on the value of the multiplier, according to whether the iterations invoke local detection:

$$\mathcal{Z} = \{k \in \mathbb{N} | \lambda_k = 0\}, \quad \mathcal{N} = \{k \in \mathbb{N} | \lambda_k > 0\}. \quad \text{eq.outer index category} \quad (3.4)$$

We first demonstrate that the ratio between the regularizer and the step size remains stable for $k \in \mathcal{N}$ as shown in (3.5) which exhibits the power of TR₊.

LEMMA 3.1. In the k -th iteration of Alg. 1, if $k \in \mathcal{N}$, then

$$d_k = -(\nabla^2 f(y_k) + \mu_k I)^{-1} \nabla f(y_k), \quad M\|d_k\| \leq \mu_k \leq 2M\|d_k\|. \quad \text{eq.output of alg2} \quad (3.5)$$

Proof. On the one hand, by our choice of (σ_k, r_k) in Line 4 of Alg. 1, we have

$$\mu_k = \sigma_k + \lambda_k \geq \sigma_k = Mr_k = M\|d_k\|.$$

On the other hand, since $\nabla^2 f(y_k) \succeq 0$, we have

$$\mu_k \|d_k\| = \|\mu_k d_k\| \leq \|(\nabla^2 f(y_k) + \mu_k I) d_k\| = \|\nabla f(y_k)\|.$$

Note that $\lambda_k > 0$ implies $\|d_k\| = r_k = \frac{1}{\sqrt{2M}} \|\nabla f(y_k)\|^{1/2}$. As a result, we have

$$\mu_k \leq \frac{\|\nabla f(y_k)\|}{\|d_k\|} = \sqrt{2M} \|\nabla f(y_k)\|^{1/2} = 2M\|d_k\|.$$

Therefore, (3.5) holds. $\square$

To maintain the flow of the complexity analysis, we temporarily assume the output of R&B of Subroutine 1 satisfies (3.5) as well, i.e., (3.5) holds for $k \in \mathcal{Z}$ (we will rigorously verify this in Section 3.2.2). Now we show that the relation (3.1) holds during the update of Alg. 1.

LEMMA 3.2. *Suppose Assumption 2.1 holds, and (3.5) holds for the output of Subroutine 1. Then we can guarantee the following for all $k \geq 0$:*

$$A_k f(x_k) \leq \phi_k^* \leq \phi_k(x) \leq A_k f(x) + \phi_0(x), \quad \forall x \in \mathbb{R}^n. \quad (3.6)$$

Proof. We conduct the proof by induction. Note that $A_0 = 0$, so (3.6) holds for $i = 0$. Suppose it also holds for $i = k$. We first check (3.1b) for $i = k + 1$:

$$\begin{aligned} \phi_{k+1}(x) &=_{(a)} \phi_k(x) + a_k (f(x_{k+1}) + \langle \nabla f(x_{k+1}), x - x_{k+1} \rangle) \\ &\leq_{(b)} A_k f(x) + \phi_0(x) + a_k (f(x_{k+1}) + \langle \nabla f(x_{k+1}), x - x_{k+1} \rangle) \\ &\leq_{(c)} A_k f(x) + \phi_0(x) + a_k f(x) \\ &=_{(d)} A_{k+1} f(x) + \phi_0(x), \end{aligned}$$

where (a) is from (3.3), (b) is from the induction hypothesis, (c) is from the convexity of f , and (d) is from (3.2). We now check (3.1a) for $i = k + 1$,

$$\begin{aligned} \phi_{k+1}^* &= \min_{x \in \mathbb{R}^n} \{ \phi_k(x) + a_k (f(x_{k+1}) + \langle \nabla f(x_{k+1}), x - x_{k+1} \rangle) \} \\ &\geq_{(a)} \min_{x \in \mathbb{R}^n} \left\{ \phi_k^* + 4M \|x - v_k\|^3 + a_k (f(x_{k+1}) + \langle \nabla f(x_{k+1}), x - x_{k+1} \rangle) \right\} \\ &\geq_{(b)} \min_{x \in \mathbb{R}^n} \left\{ A_k f(x_k) + 4M \|x - v_k\|^3 + a_k (f(x_{k+1}) + \langle \nabla f(x_{k+1}), x - x_{k+1} \rangle) \right\} \\ &\geq_{(c)} \min_{x \in \mathbb{R}^n} \left\{ A_k f(x_{k+1}) + A_k \langle \nabla f(x_{k+1}), x_k - x_{k+1} \rangle + 4M \|x - v_k\|^3 + a_k (f(x_{k+1}) + \langle \nabla f(x_{k+1}), x - x_{k+1} \rangle) \right\} \\ &=_{(d)} A_{k+1} f(x_{k+1}) + A_{k+1} \left\langle \nabla f(x_{k+1}), \frac{A_k}{A_{k+1}} x_k + \frac{a_k}{A_{k+1}} v_k - x_{k+1} \right\rangle - \frac{a_k^{3/2}}{3\sqrt{3} \cdot \sqrt{M}} \|\nabla f(x_{k+1})\|^{3/2} \\ &=_{(e)} A_{k+1} f(x_{k+1}) + A_{k+1} \langle \nabla f(x_{k+1}), y_k - x_{k+1} \rangle - \frac{a_k^{3/2}}{3\sqrt{3} \cdot \sqrt{M}} \|\nabla f(x_{k+1})\|^{3/2} \\ &= A_{k+1} f(x_{k+1}) + A_{k+1} \left(\langle \nabla f(x_{k+1}), y_k - x_{k+1} \rangle - A_{k+1}^{-1} \cdot \frac{a_k^{3/2}}{3\sqrt{3} \cdot \sqrt{M}} \|\nabla f(x_{k+1})\|^{3/2} \right) \\ &\geq_{(f)} A_{k+1} f(x_{k+1}) + A_{k+1} \left(\langle \nabla f(x_{k+1}), y_k - x_{k+1} \rangle - \frac{\sqrt{6}}{6\sqrt{M}} \|\nabla f(x_{k+1})\|^{3/2} \right). \end{aligned}$$

In the above analysis, (a) is from (A.3), (b) is from the induction hypothesis, (c) is from the convexity of f , (d) is from

$$\begin{aligned} &\min_x \left\{ A_k f(x_{k+1}) + A_k \langle \nabla f(x_{k+1}), x_k - x_{k+1} \rangle + 4M \|x - v_k\|^3 + a_k (f(x_{k+1}) + \langle \nabla f(x_{k+1}), (x - v_k) + v_k - x_{k+1} \rangle) \right\} \\ &= A_{k+1} f(x_{k+1}) + \langle \nabla f(x_{k+1}), A_k x_k - A_{k+1} x_{k+1} + a_k v_k \rangle + \min_y \{ 4M \|y\|^3 + \langle a_k \nabla f(x_{k+1}), y \rangle \} \\ &= A_{k+1} f(x_{k+1}) + A_{k+1} \left\langle \nabla f(x_{k+1}), \frac{A_k}{A_{k+1}} x_k + \frac{a_k}{A_{k+1}} v_k - x_{k+1} \right\rangle - \frac{1}{3\sqrt{3}M} \|a_k \nabla f(x_{k+1})\|^{3/2}; \end{aligned}$$

(e) is from (3.2), and (f) is from (A.1). Now it reduces to proving

$$\langle \nabla f(x_{k+1}), y_k - x_{k+1} \rangle - \frac{\sqrt{6}}{6\sqrt{M}} \|\nabla f(x_{k+1})\|^{3/2} \geq 0.$$

Using Lemma 3.1,

$$M \|d_k\| \leq \mu_k \leq 2M \|d_k\|.$$

As a result of (2.9), it holds that

$$\langle \nabla f(x_{k+1}), y_k - x_{k+1} \rangle - \frac{\sqrt{6}}{6\sqrt{M}} \|\nabla f(x_{k+1})\|^{3/2} \geq 0.$$

Thus the proof is complete. $\square$

We have the following global iteration complexity result of Alg. 1.

THEOREM 3.1. *Suppose Assumption 2.1 and Assumption 2.2 hold, and suppose that the ratio condition required by Lemma 3.2 is satisfied at every non-terminated iteration. Then for every generated iterate x_k , $k \geq 1$, we have*

$$f(x_k) - f^* \leq \frac{48M\|x_0 - x^*\|^3}{k(k+1)(k+2)}.$$

Proof. Let $R = \|x^* - x_0\|$; by Lemma 3.2 with $x = x^*$, for every $k \geq 1$, $f(x_k) - f^* \leq \phi_0(x^*)/A_k = 8MR^3/(k(k+1)(k+2)/6) = 48MR^3/[k(k+1)(k+2)]$. $\square$

COROLLARY 3.1. *Suppose Assumption 2.1 and Assumption 2.2 hold, and suppose that the ratio condition required by Lemma 3.2 is satisfied at every non-terminated iteration. Then it takes Alg. 1 $O\left(\left(\frac{MD_0^3}{\epsilon_f}\right)^{1/3}\right)$ iterations to find ϵ_f -function-value solutions as in Definition 2.1.*

3.2.2. Oracle complexity of Subroutine 1 It remains to analyze how many oracles are needed in R&B to find a direction that satisfies (3.5). We will show that the number of TR_+ oracles needed is bounded above by $O(\log(1/\epsilon_g))$. As the argument is quite technical, some of the proofs are postponed to Appendix B.

We begin with the following property of the iterates y_k with $k \in \mathcal{Z}$.

LEMMA 3.3. *Suppose Assumption 2.3 hold, for $k \in \mathcal{Z}$, we have*

$$\|\nabla f(y_k)\| \leq \frac{2\kappa_H^2}{M}. \quad (3.7)$$

Proof. Note that when $k \in \mathcal{Z}$, we have $\lambda_k = 0$, also from Line 4 hence the following hold

$$\left(\nabla^2 f(y_k) + \frac{\sqrt{2M}}{2} \|\nabla f(y_k)\|^{1/2} I \right) d_+ = -\nabla f(y_k), \quad \|d_+\| \leq \frac{1}{\sqrt{2M}} \|\nabla f(y_k)\|^{1/2}.$$

Therefore, by the triangle inequality, it holds that

$$\|\nabla f(y_k)\| \leq \|\nabla^2 f(y_k) d_+\| + \frac{\sqrt{2M}}{2} \|\nabla f(y_k)\|^{1/2} \|d_+\| \leq \|\nabla^2 f(y_k) d_+\| + \frac{1}{2} \|\nabla f(y_k)\|.$$

By Assumption 2.3, and rearranging the terms we have

$$\kappa_H \cdot \frac{1}{\sqrt{2M}} \|\nabla f(y_k)\|^{1/2} \geq \kappa_H \|d_+\| \geq \|\nabla^2 f(y_k) d_+\| \geq \frac{1}{2} \|\nabla f(y_k)\|.$$

By rearranging terms, we obtain (3.7). $\square$

COROLLARY 3.2. *Suppose Assumption 2.1, Assumption 2.3, and the "if" condition $\|d_+\| \leq \min\{\frac{\epsilon_g}{2\kappa_H}, \sqrt{\frac{\epsilon_g}{M}}\}$ at line 2 of Subroutine 1 holds, then we have*

$$\|\nabla f(y + d_+)\| \leq \epsilon_g.$$

Proof. From Lemma 2.2, we have

$$\|\nabla f(y + d_+)\| \leq \frac{M}{2} \|d_+\|^2 + \frac{\sqrt{2M}}{2} \|\nabla f(y)\|^{1/2} \|d_+\| \leq \frac{M}{2} \times \frac{\epsilon_g}{M} + \frac{\sqrt{2M}}{2} \times \frac{\sqrt{2}\kappa_H}{\sqrt{M}} \times \frac{\epsilon_g}{2\kappa_H} \leq \epsilon_g,$$

where the first in equality is also from $\lambda = 0$ and Line 4. $\square$

This implies that, in the subsequent analysis, we may assume that $\|d_+\| > \min\{\frac{\epsilon_g}{2\kappa_H}, \sqrt{\frac{\epsilon_g}{M}}\}$. Otherwise, an ϵ_g -stationary-solution has already been found.

Now we focus on the oracle complexity of R&B, i.e., the complexity of bisection over radius r on the interval $[r_-, r_+]$ to make (3.5) hold. For simplicity, we omit the subscript k in the analysis of the bisection procedure. In each step of the bisection, the update $(d, \mu) = \text{TR}_+(y, 0, r)$ with $r \in [r_-, r_+]$ defines the correspondence between r and μ over the interval $[r_-, r_+]$ and $[\mu_-, \mu_+]$, where μ_- and μ_+ have the following value due to the mechanism of Alg. 1 and Subroutine 1

$$\mu_- = M\|d_+\|, \quad \mu_+ = \frac{\sqrt{2M}}{2}\|\nabla f(y)\|^{1/2}. \quad \text{eq.value_mu} \quad (3.8)$$

We show that this correspondence is one-to-one.

LEMMA 3.4. *For any $r \in [r_-, r_+]$, there exists a unique $\mu \in [\mu_-, \mu_+]$ such that* lem.one-to-one correspondence

$$r = \|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\| := r_y(\mu),$$

establishing a one-to-one correspondence between r and μ . In particular, we may write $\mu = r_y^{-1}(r)$, and the correspondence of endpoints satisfies

$$r_- = r_y(\mu_+), \quad r_+ = r_y(\mu_-).$$

Proof. The endpoints correspondence between $r_- \rightarrow \mu_+$ and $r_+ \rightarrow \mu_-$ follows from line 16 of Subroutine 1 and (3.8).

For any r , the existence of such μ follows from $(d, \mu) = \text{TR}_+(y, 0, r)$ and (2.10c). For uniqueness, noting $\nabla^2 f(y) + \mu I \succ 0$ for any $\mu \in [\mu_-, \mu_+]$, we can apply the eigenvalue decomposition $\nabla^2 f(y) = V\Lambda V^\top$ with $\Lambda = \text{diag}(\zeta_1, \dots, \zeta_n)$ and V orthogonal. Then

$$r = \|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\| = \sqrt{\sum_{i=1}^n \frac{\beta_i^2}{(\zeta_i + \mu)^2}},$$

where $\beta_i = \nabla f(y)^\top v_i$. Since $\zeta_i \geq 0$ and $\mu > 0$, every term in the summation is continuous and strictly decreasing in μ , and thus $r_y(\mu)$ is continuous and strictly decreasing on $[\mu_-, \mu_+]$. Consequently, for any $r \in [r_-, r_+]$, there exists a unique $\mu \in [\mu_-, \mu_+]$ such that

$$r = r_y(\mu) = \|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\|.$$

Hence, the correspondence between r and μ is one-to-one, establishing the desired relation $\mu = r_y^{-1}(r)$. $\square$

Using this one-to-one correspondence, when we perform bisection on r over the interval $[r_-, r_+]$, we also implicitly perform bisection on μ over $[\mu_-, \mu_+]$. The logic of our analyses is to first identify the target interval of μ and then use the one-to-one correspondence again to transfer it to the target interval of r . This observation motivates us to introduce the following auxiliary functions of μ .

$$g_y(\mu) = \frac{\mu}{r_y(\mu)} = \frac{\mu}{\|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\|}. \quad \text{eq.auxiliary function for bisection} \quad (3.9)$$

The goal of bisection is to locate the radius r such that the corresponding μ as defined in Lemma 3.4 satisfies

$$M \leq g_y(\mu) \leq 2M,$$

as we have established in Lemma 3.1 for $k \in \mathcal{N}$.

The following lemma validates the choices of the bracketing points at line 16.

LEMMA 3.5. *In R&B of Subroutine 1, when the bisection begins, we have* lem.valid bracketing

$$g_y(\mu_-) < M, \quad g_y(\mu_+) > 2M.$$

Proof. From line 13 in Subroutine 1, we have $g_y(\mu_+) = \frac{\mu_+}{\|d_+\|} > 2M$ whenever Subroutine 1 enters bisection procedure.

Denote $d_- = -(\nabla^2 f(y) + \mu_- I)^{-1} \nabla f(y)$. Since $\mu_- < \mu_+$, then

$$\|d_-\| = \|(\nabla^2 f(y) + \mu_- I)^{-1} \nabla f(y)\| > \|(\nabla^2 f(y) + \mu_+ I)^{-1} \nabla f(y)\| = \|d_+\|,$$

therefore $g_y(\mu_-) = \frac{\mu_-}{\|d_-\|} = \frac{M\|d_+\|}{\|d_-\|} < M$. $\square$

Next, we identify the length of the target interval for μ .

lem.bound target interval

LEMMA 3.6. *There exist μ_l, μ_u with $\mu_- < \mu_l < \mu_u < \mu_+$ such that*

$$g_y(\mu_l) = M, \quad g_y(\mu_u) = \frac{M}{2}, \quad \text{eq.target interval bracketing points} \quad (3.10)$$

and for all $\mu \in [\mu_l, \mu_u]$, we have

$$M \leq g_y(\mu) \leq 2M.$$

The length of the target interval $[\mu_l, \mu_u]$ satisfies

$$\mu_u - \mu_l \geq \frac{M\|(\nabla^2 f(y) + \mu_u I)^{-1} \nabla f(y)\|}{2}. \quad \text{eq.bound target interval} \quad (3.11)$$

After using the one-to-one correspondence to transfer the target interval to the one with respect to r , the complexity of R&B is bounded as follows. The details can be found in the Appendix B.

lem.bound initial interval

LEMMA 3.7. *Suppose Assumption 2.1 and Assumption 2.3 hold. At the k -th iteration of Alg. 1, the oracle complexity of the bisection search procedure is*

$$O\left(\log \frac{\kappa_H + \mu_+}{\epsilon_g}\right). \quad \text{eq.complexity bisection} \quad (3.12)$$

Till now, we have established that the oracle complexity of Subroutine 1 reduces to $\max\{O\left(\log \frac{\kappa_H + \mu_+}{\epsilon_g}\right), O\left(\log \frac{\|\nabla f(y)\|}{\epsilon_g}\right)\}$. Since $\mu_+ = \frac{\sqrt{2M}}{2}\|\nabla f(y)\|^{1/2}$ and we already established the boundedness for the gradient norm in Lemma 3.3, we can derive the total complexity of Alg. 1,

thm.total complexity newton

THEOREM 3.2. *Suppose Assumption 2.1, Assumption 2.2 and Assumption 2.3 hold, it takes at most*

$$O\left(\epsilon_f^{-1/3} \log(1/\epsilon_g)\right) \quad \text{eq.total complexity newton} \quad (3.13)$$

TR_+ oracles for Alg. 1 to find an (ϵ_f, ϵ_g) -approximate solution that satisfies Definition 2.1.

Proof. This result is a direct combination of Theorem 3.1 with Lemma 3.7 and Lemma 3.3. $\square$

sec.local

3.3. Local convergence rate of Alg. 1 In this section, we analyze the local convergence rate of Alg. 1. Compared to other accelerated SOMs, Alg. 1 shows a quadratic local convergence in favor of the local detection mechanism. The quadratic local convergence is essential to find high-accuracy approximate solutions. First, we make the following standard assumption for local convergence analysis.

assm.local strongly convex

ASSUMPTION 3.1. *Suppose problem (1.1) has a unique solution x^* , which satisfies the second-order sufficient optimality condition*

$$\nabla f(x^*) = 0, \quad \nabla^2 f(x^*) \succeq \nu I \quad \text{eq.second-order suff cond} \quad (3.14)$$

for some $\nu > 0$.

If the above assumption holds, it is a well-known result that Newton's method converges quadratically if initialized in a small region that contains the optimum.

lem.quadratic point

LEMMA 3.8 (Theorem 1.2.5, [47]). *Suppose Assumption 2.1 and Assumption 3.1 hold and $\{z_i\}$ is the sequence generated by Newton's method. Denoting $R_i = \|z_i - x^*\|$, when the initial point z_0 is in the region:*

$$\mathcal{LQ}_p = \{x \in \mathbb{R}^n \mid \|x - x^*\| < \frac{2\nu}{3M}\}, \quad \text{eq.local quadratic region} \quad (3.15)$$

then $z_i \in \mathcal{LQ}_p$ for all i and it converges quadratically to x^* :

$$R_{i+1} < \frac{3M}{2\nu} R_i^2 < R_i. \quad \text{eq.quadratic convergence} \quad (3.16)$$

The subscript p is mnemonic for point-wise. Similar to the standard quadratic convergence region discussed above, the gradient norm also exhibits quadratic convergence when the initial point is sufficiently close to x^* (we will specify the value of r in the following lemma later). To establish this result, we first present the following technical lemma, which will be central to the subsequent analysis.

LEMMA 3.9 (Corollary 1.2.2, [47]). Suppose Assumption 2.1 and Assumption 3.1 hold, for any x with $\|x - x^*\| \leq r$, we have

$$\nabla^2 f(x^*) - MrI \preceq \nabla^2 f(x) \preceq \nabla^2 f(x^*) + MrI. \quad \text{eq.psd region} \quad (3.17)$$

Therefore, when $x \in \mathcal{LQ}_p$, we have

$$\nabla^2 f(x) \succ \frac{\nu}{3}I. \quad \text{eq.psd xk} \quad (3.18)$$

Now we present the region of quadratic convergence $\mathcal{LQ}_g$ for the norm of the gradient, the subscript g is mnemonic for gradient-wise.

LEMMA 3.10. Suppose Assumption 2.1 and Assumption 3.1 hold and $\{z_i\}$ is the sequence generated by Newton's method. When the initial point z_0 is in the region:

$$\mathcal{LQ}_g = \{x \in \mathbb{R}^n \mid \|x - x^*\| < \frac{2\nu}{3M}, \|\nabla f(x)\| < \frac{2\nu^2}{9M}\}, \quad \text{eq.loc 2v^2 quadratic region gradient} \quad (3.19)$$

then $z_i \in \mathcal{LQ}_g$ for all i and the norm of the gradient $\|\nabla f(z_i)\|$ converges quadratically to 0:

$$\|\nabla f(z_{i+1})\| < \frac{9M}{2\nu^2} \|\nabla f(z_i)\|^2 < \|\nabla f(z_i)\|. \quad \text{eq.quadratic convergence gradient} \quad (3.20)$$

Proof. We prove by induction. Suppose $z_i \in \mathcal{LQ}_g$. From (2.5a), we have

$$\|\nabla f(z_{i+1}) - \nabla f(z_i) - \nabla^2 f(z_i)(z_{i+1} - z_i)\| \leq \frac{M}{2} \|z_{i+1} - z_i\|^2.$$

Noting that $z_{i+1} = z_i - \nabla^2 f(z_i)^{-1} \nabla f(z_i)$, we have

$$\|\nabla f(z_{i+1})\| \leq \frac{M}{2} \|\nabla^2 f(z_i)^{-1} \nabla f(z_i)\|^2 \leq \frac{M}{2} \|\nabla^2 f(z_i)^{-1}\|^2 \|\nabla f(z_i)\|^2 < \frac{9M}{2\nu^2} \|\nabla f(z_i)\|^2 < \|\nabla f(z_i)\|.$$

The third inequality is from $R_i < \frac{2\nu}{3M}$ and (3.18). The last inequality is from the fact $z_i \in \mathcal{LQ}_g$. $R_{i+1} \leq \frac{2\nu}{3M}$ follows from Lemma 3.8. Therefore, $z_{i+1} \in \mathcal{LQ}_g$. $\square$

As a result, we know how many iterations are needed to find a point with $\|\nabla f(z_i)\| \leq \epsilon_g$ if $z_0 \in \mathcal{LQ}_g$ and $\{z_i\}$ is generated by Newton's method initialized at z_0 .

COROLLARY 3.3. Suppose Assumption 2.1 and Assumption 3.1 hold. Let $0 < \epsilon_g < \frac{1}{C(M, \nu)}$, $z_0 \in \mathcal{LQ}_g$ and $\{z_i\}$ generated by Newton's method. Then when

$$i \geq \left\lceil \log \left(\frac{\ln \left(\frac{1}{C(M, \nu) \epsilon_g} \right)}{\ln \left(\frac{1}{\eta_0} \right)} \right) \right\rceil \quad \text{with} \quad C(M, \nu) = \frac{9M}{2\nu^2}, \quad \eta_0 = \frac{\|\nabla f(z_0)\|}{C(M, \nu)} < 1, \quad \text{eq.number of iterations newton} \quad (3.21)$$

we have $\|\nabla f(z_i)\| \leq \epsilon_g$.

Proof. From (3.20) we have

$$\|\nabla f(z_i)\| \leq C(M, \nu) \|\nabla f(z_{i-1})\|^2 \leq C(M, \nu)^{2^{i-1}} \|\nabla f(z_0)\|^{2^i} \leq \frac{(C(M, \nu) \|\nabla f(z_0)\|)^{2^i}}{C(M, \nu)}.$$

Therefore, when i hits the threshold defined in (3.21), we have $\|\nabla f(z_i)\| \leq \epsilon_g$. $\square$

After all these preparations, we show that Subroutine 1 can guarantee that we can find an (ϵ_f, ϵ_g) -approximate solution with a local quadratic rate. The proof consists of the following three steps:

1. After constant number of iterations of Alg. 1, $y_k \in \mathcal{LQ}_g$;
2. Whenever $y_k \in \mathcal{LQ}_g$, $\lambda_k = 0$ in Alg. 1;
3. Once $y_k \in \mathcal{LQ}_g$, the Local Diving procedure in Subroutine 1 converges quadratically with ET=True.

First, we show the first step, i.e., y_k will finally enter the region $\mathcal{LQ}_g$.

LEMMA 3.11. Suppose Assumptions 2.1, 2.3, and 3.1 hold. Let $R = \|x_0 - x^*\|$. Then, whenever the k -th iterate of Alg. 1 is generated, if

$$k \geq \left\lceil 36 \frac{\kappa_H MR}{\nu^2} \right\rceil,$$

we have $y_k \in \mathcal{LQ}_g$.

The proof of the above lemma can be found in Section B. Next, we prove the second step: in Alg. 1, when $y_k \in \mathcal{LQ}_g$, we must have $\lambda_k = 0$.

LEMMA 3.12. *Suppose Assumption 2.1 and Assumption 3.1 hold. In Alg. 1, if $y_k \in \mathcal{LQ}_g$, then $\lambda_k = 0$.* lem.detection of local quadratic

Proof. From (2.10b), to prove $\lambda_k = 0$, it suffices to prove that the step d_k lies in the trust region. **First from Line 4 in Alg. 1**

$$\begin{aligned} \|(\nabla^2 f(y_k) + \sigma_k I)^{-1} \nabla f(y_k)\| &= \left\| \left(\nabla^2 f(y_k) + \frac{\sqrt{2M}}{2} \|\nabla f(y_k)\|^{1/2} I \right)^{-1} \nabla f(y_k) \right\| \\ &\leq \left\| \left(\nabla^2 f(y_k) + \frac{\sqrt{2M}}{2} \|\nabla f(y_k)\|^{1/2} I \right)^{-1} \right\| \|\nabla f(y_k)\| \end{aligned}$$

Note $\nabla^2 f(y_k) \succ \frac{\nu}{3}$ as analyzed in (3.18).

$$\begin{aligned} \|(\nabla^2 f(y_k) + \sigma_k I)^{-1} \nabla f(y_k)\| &\leq \frac{\|\nabla f(y_k)\|}{\frac{\nu}{3} + \frac{\sqrt{2M}}{2} \|\nabla f(y_k)\|^{1/2}} \\ &< \frac{\|\nabla f(y_k)\|}{\frac{\sqrt{2M}}{2} \|\nabla f(y_k)\|^{1/2} + \frac{\sqrt{2M}}{2} \|\nabla f(y_k)\|^{1/2}} = \frac{\|\nabla f(y_k)\|^{1/2}}{\sqrt{2M}} = r_k. \end{aligned}$$

The second line is because of the definition of $\mathcal{LQ}_g$ in (3.19). Therefore, we conclude that the TR constraint is inactive and $\lambda_k = 0$. $\square$

Last, we show the third step: when the tolerance ϵ_g is small enough, Alg. 1 will be early terminated, i.e., we have ET=True when y_k enters $\mathcal{LQ}_g$.

THEOREM 3.3. *Suppose Assumption 2.1 and Assumption 3.1 hold, then there exists $\epsilon^* > 0$ such that when* thm.local quadratic convergence

$$0 < \epsilon_g < \min\{\epsilon^*, \frac{1}{C(M, \nu)}\}, \quad \text{eq.regularity epsilon} \quad (3.22)$$

the ϵ_g -stationary-solutions will be found in $O(\log \log(1/\epsilon_g))$ iterations when y_k enters $\mathcal{LQ}_g$, and ET=True in Alg. 1.

Proof. We first consider the case that ET=True at line 3 of Subroutine 1, then $y_k + d_k$ is already an ϵ_g -stationary-solution (see Corollary 3.2) and there is nothing to prove.

Next, we consider the case Subroutine 1 enters local diving. We know from (3.18) that when $y_k \in \mathcal{LQ}_g$, $\|\nabla f(y_k)\| \leq \frac{1}{C(M, \nu)}$, then from Lemma B.2 (see Appendix B) we know that local diving will find ϵ_g -stationary-solutions within the iteration number defined in (3.21), which is strictly less than the maximal number of iteration we defined at line 6, i.e., $\lceil \log \frac{\|\nabla f(y)\|}{\epsilon_g} \rceil$. $\square$

4. Variant II: Pushing Global Efficiency to the Limit We now turn to another accelerated TR method that incorporates the acceleration framework of Monteiro and Svaiter [42]. This method achieves a near-optimal global oracle complexity of $\tilde{O}(\epsilon_f^{-2/7})$ for finding (ϵ_f, ϵ_g) -approximate solutions. However, this global efficiency comes at the cost of losing local quadratic convergence. We refer to this approach as the Accelerated Trust-Region Extragradient Method (Alg. 2). sec.second variant

4.1. Algorithm design We begin by outlining the acceleration framework in Alg. 2. The main technical ingredient is to integrate the TR_+ oracle into a modified version of the framework proposed by Monteiro and Svaiter [42], providing another eligible oracle choice besides the cubic regularization oracle in the previous literature [7].

At the core of Alg. 2 lies an implicit search procedure (Subroutine 2, we still call it R&B for short) along the curve **probably replace y_k with $y_k(\sigma)$** :

$$y_k(\sigma) = \frac{A_k}{A_k + a_k(\sigma)} x_k + \frac{a_k(\sigma)}{A_k + a_k(\sigma)} v_k \quad \text{eq.update } y_k \quad (4.1)$$

$$a_k(\sigma) = \frac{1 + \sqrt{1 + 4A_k\sigma}}{2\sigma}. \quad \text{eq.update_ak (4.2)}$$

In the whole process, R&B (Subroutine 2) determines the primal regularization σ_k , and thus $a_k(\sigma_k)$ and the extrapolation point $y_k(\sigma_k)$, at which TR_+ is called. In R&B (Subroutine 2), the search task initializes

Algorithm 2 Accelerated Trust-Region Extragradient Method (Alg. 2)

```

1: input:  $x_0 = v_0 \in \mathbb{R}^n$ ,  $A_0 = 0$ , R&B search threshold  $\theta > 1$ ,  $0 < \eta < 1$ , damping parameter  $\gamma \leq \frac{1}{\theta}$ ,
   stationarity tolerance  $\epsilon_g > 0$ 
2: for  $k = 0, 1, 2, \dots$  do
3:    $(x_{k+1}, \sigma_k, \text{ET}) = \text{R\&B}(x_k, v_k, A_k, \eta, \theta, \epsilon_g)$ 
4:   if ET then
5:     terminate and output  $x_{k+1}$ 
6:   end if
7:    $a_k = a_k(\sigma_k)$ 
8:    $v_{k+1} = v_k - \gamma a_k \nabla f(x_{k+1})$ 
9:    $A_{k+1} = A_k + a_k$ 
10: end for

```

Subroutine 2 Ratio Bracketing and Bisection (R&B) for Alg. 2

```

1: input:  $x, v \in \mathbb{R}^n$ ,  $A, 0 < \eta < 1, \theta > 1, \epsilon_g > 0$ ;
2: # Bracket points
3: Set  $\sigma_- = \sqrt{\frac{2M\epsilon_g}{1+2\theta}}$ ,  $\sigma_+ = \sqrt{\frac{MG_0}{\eta}}$ ,  $(\lambda_-, d_-) = \text{TR}_+(y(\sigma_-), \sigma_-, \frac{1}{M}\sigma_-)$ 
4: if  $\lambda_- \leq (\theta - 1)\sigma_-$  then
5:   Output  $y(\sigma_-) + d_-, \sigma_-, \text{ET}=\text{True}$ 
6: else
7:   # Perform bisection over  $\sigma \in [\sigma_-, \sigma_+]$ 
8:   while i)  $\lambda = 0$  and  $\|d\| < \frac{\eta}{M}\sigma$ , or ii)  $\lambda > (\theta - 1)\sigma$  do
9:      $\sigma = \frac{\sigma_- + \sigma_+}{2}$ 
10:     $(\lambda, d) = \text{TR}_+(y(\sigma), \sigma, \frac{1}{M}\sigma)$ 
11:    if i) holds then
12:       $\sigma_+ = \sigma$ 
13:    else if ii) holds then
14:       $\sigma_- = \sigma$ 
15:    end if
16:  end while
17:  output  $y(\sigma) + d, \sigma, \text{ET}=\text{False}$ 
18: end if

```

the bracketing points at line 3, where

$$G_0 = \max \left\{ \|\nabla f(x)\| \mid \|x - x^*\| \leq \left(\frac{4}{\sqrt{3}\gamma} + 1 \right) D_0 \right\}, \quad D_0 = \|x_0 - x^*\|.$$

At line 8, we proceed bisection until σ_k has been located such that

$$0 \leq \lambda_k \leq (\theta - 1)\sigma_k, \quad \|d_k\| \geq \frac{\eta}{M}\sigma_k, \quad \text{eq.relation_lambda_sigma (4.3)}$$

where θ and η are prescribed thresholds for the regularization term and the step size, respectively. They will work together with the damping parameter γ in Alg. 2 to guarantee acceleration for TR_+ oracle. Inequalities (4.3) serves a similar purpose as (3.1) to Alg. 1, which we will further clarify in Lemma 4.4.

Similar to Alg. 1, there are two paths to terminate Alg. 2. The first path occurs when the flag $\text{ET} = \text{True}$, indicating early termination (see line 5). This happens when an ϵ_g -stationary-solution is found in R&B (Subroutine 2). The second path arises if Alg. 2 is not early terminated. In this case, the step d_k satisfies (4.3) for each k , and consequently, the estimating sequence guarantees that Alg. 2 outputs an ϵ_f -function-value solution.

4.2. Global convergence in function value The proof sketch is similar to the one in the previous section: we first establish *iteration complexity* then move on to *oracle complexity*.

4.2.1. Iteration complexity of Alg. 2 We first show that Alg. 2 can terminate early in R&B (Subroutine 2).

LEMMA 4.1. *Suppose Assumption 2.1 holds and Subroutine 2 outputs $\text{ET} = \text{True}$ at line 5, we have*

$$\|\nabla f(y(\sigma_-) + d_-)\| \leq \epsilon_g.$$

Proof. From Lemma 2.1, we have

$$\|\nabla f(y(\sigma_-) + d_-) - \nabla f(y(\sigma_-)) - \nabla^2 f(y(\sigma_-))d_-\| \leq \frac{M}{2}\|d_-\|^2.$$

By triangle inequality we have

$$\begin{aligned} \|\nabla f(y(\sigma_-) + d_-)\| &\leq \frac{M}{2}\|d_-\|^2 + \|\nabla f(y(\sigma_-)) + \nabla^2 f(y(\sigma_-))d_-\| \\ &=_{(a)} \frac{M}{2}\|d_-\|^2 + (\sigma_- + \lambda_-)\|d_-\| \\ &\leq_{(b)} \frac{1+2\theta}{2M}\sigma_-^2 \leq_{(c)} \epsilon_g, \end{aligned}$$

where (a) is from (2.10c), (b) is from Line 3 and thus $\|d_-\| \leq \frac{1}{M}\sigma_-$, $\sigma_- + \lambda_- \leq \theta\sigma_-$, and (c) is from Line 3.

□

Since we focus on the worst-case global oracle complexity, we assume that early termination does not occur; that is, the “If” route at line 4 of R&B (Subroutine 2) is never triggered. Similar to previous analyses, we will show that Alg. 2 takes $O(\epsilon_f^{-2/7})$ to find an ϵ_f -function-valueapproxim solution satisfying Definition 2.1.

First, we assume the bisection procedure is valid in the sense that (4.3) holds for each iteration (which will be verified in Section 4.2.2). As a result, the following recursive relation holds.

LEMMA 4.2. *Suppose Assumption 2.1, Assumption 2.2 hold, and the output of R&B (Subroutine 2) satisfies (4.3), i.e.,*

$$0 \leq \lambda_k \leq (\theta - 1)\sigma_k, \quad \|d_k\| \geq \frac{\eta}{M}\sigma_k,$$

then in the k -th iteration of Alg. 2, the following holds:

$$\frac{1}{2}\|v_{k+1} - x^*\|^2 + \gamma A_{k+1}(f(x_{k+1}) - f^*) + \frac{3\gamma A_{k+1}\sigma_k}{8}\|d_k\|^2 \leq \frac{1}{2}\|v_k - x^*\|^2 + \gamma A_k(f(x_k) - f^*). \quad (4.4)$$

Further, if the output of R&B (Subroutine 2) satisfies (4.3) for all iteration i with $0 \leq i \leq k$, then

$$\gamma A_{k+1}(f(x_{k+1}) - f^*) + \frac{1}{2}\|v_{k+1} - x^*\|^2 + B_{k+1} \leq \frac{1}{2}\|v_0 - x^*\|^2, \quad (4.5)$$

where $B_{k+1} = \frac{3\gamma M}{8} \sum_{i=0}^k A_{i+1}\|d_i\|^3$.

Proof. Note the way we update v_{k+1} in Alg. 2.

$$\|v_{k+1} - x^*\|^2 = \|v_k - x^* - \gamma a_k \nabla f(x_{k+1})\|^2$$

$$\begin{aligned}
&= \|v_k - x^*\|^2 + \gamma^2 a_k^2 \|\nabla f(x_{k+1})\|^2 - 2\gamma a_k \langle \nabla f(x_{k+1}), v_k - x^* \rangle \\
&= \|v_k - x^*\|^2 + \gamma^2 a_k^2 \|\nabla f(x_{k+1})\|^2 - 2\gamma \langle \nabla f(x_{k+1}), (A_k + a_k)y_k - A_k x_k - a_k x^* \rangle \\
&= \|v_k - x^*\|^2 + \gamma^2 a_k^2 \|\nabla f(x_{k+1})\|^2 \\
&\quad - 2\gamma \langle \nabla f(x_{k+1}), (A_k + a_k)(y_k - x_{k+1}) + A_k(x_{k+1} - x_k) + a_k(x_{k+1} - x^*) \rangle \\
&\leq \|v_k - x^*\|^2 + \gamma^2 a_k^2 \|\nabla f(x_{k+1})\|^2 - 2\gamma (A_k + a_k) \langle \nabla f(x_{k+1}), y_k - x_{k+1} \rangle \\
&\quad - 2\gamma A_k (f(x_{k+1}) - f(x_k)) - 2\gamma a_k (f(x_{k+1}) - f^*).
\end{aligned}$$

The third line comes from (4.1). The last line is from the convexity of f . Using $A_{k+1} = A_k + a_k$ as in Alg. 2 and rearranging terms gives

$$\begin{aligned}
&\|v_{k+1} - x^*\|^2 + 2\gamma A_{k+1} (f(x_{k+1}) - f^*) \\
&\leq \|v_k - x^*\|^2 + 2\gamma A_k (f(x_k) - f^*) + \gamma^2 a_k^2 \|\nabla f(x_{k+1})\|^2 - 2\gamma (A_k + a_k) \langle \nabla f(x_{k+1}), y_k - x_{k+1} \rangle \\
&\leq \|v_k - x^*\|^2 + 2\gamma A_k (f(x_k) - f^*) + \gamma^2 a_k^2 \|\nabla f(x_{k+1})\|^2 - \gamma (A_k + a_k) \left(\frac{\|\nabla f(x_{k+1})\|^2}{\sigma_k + \lambda_k} + \frac{3(\sigma_k + \lambda_k)}{4} \|d_k\|^2 \right) \\
&\leq \|v_k - x^*\|^2 + 2\gamma A_k (f(x_k) - f^*) + \left(\gamma^2 a_k^2 - \frac{\gamma(A_k + a_k)}{\theta \sigma_k} \right) \|\nabla f(x_{k+1})\|^2 - \frac{3\gamma(A_k + a_k)(\sigma_k + \lambda_k)}{4} \|d_k\|^2.
\end{aligned}$$

The second inequality is from whenever we call the oracle at Line 3 and 10, we have $\sigma \geq M\|d\|$, hence we can apply (2.8). The third inequality is from (4.3). Note that $\gamma \leq \frac{1}{\theta}$ and (4.2) hold. We have

$$\gamma^2 a_k^2 - \frac{\gamma(A_k + a_k)}{\theta \sigma_k} = \gamma \left(\gamma a_k^2 - \frac{A_k + a_k}{\theta \sigma_k} \right) \leq \frac{\gamma}{\theta} \left(a_k^2 - \frac{A_k + a_k}{\sigma_k} \right) = 0,$$

which gives (4.4). Also, because $\sigma_k \geq M\|d_k\|$ as stated,

$$\frac{1}{2} \|v_{k+1} - x^*\|^2 + \gamma A_{k+1} (f(x_{k+1}) - f^*) + \frac{3\gamma M A_{k+1}}{8} \|d_k\|^3 \leq \frac{1}{2} \|v_k - x^*\|^2 + \gamma A_k (f(x_k) - f^*).$$

Summing the above inequality over $i = 0, 1, \dots, k$, we obtain

$$\sum_{i=0}^k \left(\frac{1}{2} \|v_{i+1} - x^*\|^2 + \gamma A_{i+1} (f(x_{i+1}) - f^*) \right) + \frac{3\gamma M}{8} \sum_{i=0}^k A_{i+1} \|d_i\|^3 \leq \sum_{i=0}^k \left(\frac{1}{2} \|v_i - x^*\|^2 + \gamma A_i (f(x_i) - f^*) \right).$$

By canceling the common terms on both sides (telescoping sum), this simplifies to

$$\frac{1}{2} \|v_{k+1} - x^*\|^2 + \gamma A_{k+1} (f(x_{k+1}) - f^*) + \frac{3\gamma M}{8} \sum_{i=0}^k A_{i+1} \|d_i\|^3 \leq \frac{1}{2} \|v_0 - x^*\|^2 + \gamma A_0 (f(x_0) - f^*).$$

Since $A_0 = 0$ by initialization, the second term on the right-hand side vanishes. Letting $B_{k+1} = \frac{3\gamma M}{8} \sum_{i=0}^k A_{i+1} \|d_i\|^3$, we obtain the desired result (4.5). $\square$

From (4.5), it is clear that the oracle complexity of Alg. 2 reduces to analyzing the growth rate of A_k , which is addressed in Monteiro and Svaiter [42, Lemma 4.2] and Nesterov [47, Lemma 4.3.5]. We provide the lower bound of A_k in the following lemma, whose proof is deferred to the Section C. lem.growth of A_k

LEMMA 4.3. For and $k \geq 1$, we have

$$A_k \geq \left(\frac{\eta}{4} \left(\frac{3\gamma}{4M^2 D_0^2} \right)^{1/3} \right)^{3/2} \left(\frac{2k+1}{3} \right)^{7/2} = \Omega \left(\frac{k^{7/2}}{M D_0} \right). \quad \text{eq.growth of } A_k \quad (4.6)$$

Hence, as a direct consequence of (4.5) and (4.6), the complexity of Alg. 2 can be summarized as follows.

THEOREM 4.1. [Theorem 4.3.2, Nesterov [47]] ^{thm.outer loop complexity} Suppose Assumption 2.1 and Assumption 2.2 hold, and suppose that (4.3) is satisfied at every non-terminated iteration. Then it takes Alg. 2 $O(MD_0^3 \epsilon_f^{-2/7})$ iterations to find ϵ_f -function-value solutions as in Definition 2.1.

4.2.2. Oracle complexity of Alg. 2 ^{sec.validty of bisection} Next, we elaborate on how R&B (Subroutine 2) safeguards (4.3), and then provide the estimate of the number of TR_+ oracles needed during this procedure. We omit

the subscripts in the analysis for the bisection for simplicity. We now define the auxiliary function in the analysis, which is bivariate in y and σ .

$$\psi(\sigma, y) := \frac{1}{\sigma} \left\| \left(\nabla^2 f(y) + \sigma I \right)^{-1} \nabla f(y) \right\|, \quad \sigma > 0, y \in \mathbb{R}^n. \quad \text{eq. line search quantity} \quad (4.7)$$

Some basic analyses on local perturbation in σ and y are deferred to Section C (see Lemma C.1 and Lemma C.2).

The below Lemma 4.4 means the analysis of auxiliary function $\psi(\sigma, y(\sigma))$ can be simplified by focusing solely on σ , reducing the search procedure into a one-dimensional problem. **The proof of Lemma 4.4 can be found in Section C.**

LEMMA 4.4. At line 8 of R&B (Subroutine 2), if case i) occurs, i.e.,

$$\lambda = 0, \quad \|d\| < \frac{\eta}{M} \sigma, \quad \text{eq. bracket small lambda} \quad (4.8)$$

then $\psi(\sigma, y(\sigma)) < \frac{\eta}{M}$. Otherwise, if case ii) occurs, i.e.,

$$\lambda > (\theta - 1)\sigma, \quad \text{eq. bracket big lambda} \quad (4.9)$$

then $\psi(\sigma, y(\sigma)) > \frac{1}{M}$. As a result, if σ satisfies

$$\frac{\eta}{M} \leq \psi(\sigma, y(\sigma)) \leq \frac{1}{M}, \quad \text{eq. reduce line search interval} \quad (4.10)$$

then (4.3) holds.

Now we can validate the choice of σ_-, σ_+ as qualified bracketing points.

LEMMA 4.5. Suppose

$$\|\nabla f(y(\sigma))\| \leq G_0, \quad \forall \sigma > 0, \quad \text{lem. bracketing points} \quad \text{eq. bounded gradient y} \quad (4.11)$$

and we let

$$\sigma_- = \sqrt{\frac{2M\epsilon_g}{1+2\theta}}, \quad \sigma_+ = \sqrt{\frac{MG_0}{\eta}}. \quad \text{eq. bracket points} \quad (4.12)$$

Then if R&B (Subroutine 2) does not early terminate, we have

$$\psi(\sigma_-, y(\sigma_-)) > \frac{1}{M}, \quad \psi(\sigma_+, y(\sigma_+)) < \frac{\eta}{M}.$$

Proof. First, we show that $\psi(\sigma_+, y(\sigma_+)) < \frac{\eta}{M}$,

$$\begin{aligned} \psi(\sigma_+, y(\sigma_+)) &= \frac{1}{\sigma_+} \left\| \left(\nabla^2 f(y(\sigma_+)) + \sigma_+ I \right)^{-1} \nabla f(y(\sigma_+)) \right\| \\ &\leq \frac{1}{\sigma_+^2} \|\nabla f(y(\sigma_+))\| \leq \frac{1}{\sigma_+^2} G_0 = \frac{\eta}{M}. \end{aligned}$$

The second line is from $\nabla^2 f(y(\sigma_+)) \succeq 0$, (4.11) and (4.12). For the other statement, from Lemma 4.1, we can conclude that if the Subroutine 2 does not early terminate, we have $\lambda_- > (\theta - 1)\sigma_-$, and from Lemma 4.4 $\psi(\sigma_-, y(\sigma_-)) > \frac{1}{M}$. $\square$

Under the bounded gradient assumption (4.11), during the bisection, we have

$$\psi_+ := \psi(\sigma_+, y(\sigma_+)) < \frac{\eta}{M}, \quad \psi_- := \psi(\sigma_-, y(\sigma_-)) > \frac{1}{M}, \quad \text{eq. bisection condition} \quad (4.13)$$

$$\sigma_- \geq \sqrt{\frac{2M\epsilon_g}{1+2\theta}}, \quad \sigma_+ \leq \sqrt{\frac{MG_0}{\eta}}. \quad \text{eq. bracket points condition} \quad (4.14)$$

These relations ensure that valid bracketing points are maintained throughout the search. Leveraging the property of the curve $\psi(\sigma, y(\sigma))$, we can then derive the complexity of R&B (Subroutine 2) as in Lemma 4.6. A formal statement and proof of this result are provided in Section C (Lemma C.5).

LEMMA 4.6. *Assume that Assumptions 2.1, Assumption 2.2 and conditions (4.11) hold. Also, assume* lem.complexity bisection

$$\|x - x^*\| \leq M_0, \|v - x^*\| \leq M_0,$$
 eq.bounded assm point main text (4.15)

for some $M_0 > 0$, then the number of TR_+ oracle calls during the bisection is $O(\log(1/\epsilon_g))$.

To derive the final oracle complexity result of Alg. 2, one final step remains: we must get rid of the boundedness assumption used in (4.11) and (4.15). **We establish this in Lemma 4.7, the proof of which can be found in Section C.**

LEMMA 4.7. *Suppose Assumption 2.1 and Assumption 2.2 hold. For every $k \geq 0$,* lem.bounded iterate

$$\|x_k - x^*\| \leq \left(\frac{4}{\sqrt{3\gamma}} + 1 \right) D_0, \quad \|v_k - x^*\| \leq D_0. \quad \text{eq.bounded iterate (4.16)}$$

As a consequence,

$$\|y_k(\sigma) - x^*\| \leq \left(\frac{4}{\sqrt{3\gamma}} + 1 \right) D_0, \quad \forall \sigma > 0, \quad \text{eq.bounded iterate y (4.17)}$$

$$\|\nabla f(y_k(\sigma))\| \leq G_0 := \left(\frac{4}{\sqrt{3\gamma}} + 1 \right) \|\nabla^2 f^*\| D_0 + \frac{M}{2} \left(\frac{4}{\sqrt{3\gamma}} + 1 \right)^2 D_0^2. \quad \text{eq.ub yk sigma (4.18)}$$

By this lemma, we know (4.11) holds, and (4.15) holds uniformly for all $k \geq 0$ with $M_0 = \left(\frac{4}{\sqrt{3\gamma}} + 1 \right) D_0$. Now we finally arrive at the final theorem of Alg. 2, as a consequence of Theorem 4.1, Lemma 4.6, and Lemma 4.7.

THEOREM 4.2. *Suppose Assumption 2.1 and Assumption 2.2 hold, it takes at most* thm.final complexity of msatr

$$O\left(\epsilon_f^{-2/7} \log(1/\epsilon_g)\right) \quad \text{eq.final complexity of msatr (4.19)}$$

TR_+ oracles for Alg. 2 to find an (ϵ_f, ϵ_g) -approximate solution as in Definition 2.1.

While the proposed algorithm attains a near-optimal global oracle complexity rate, it fails to balance the global guarantees and the local efficiency. The reason behind is that the primal regularizer σ must be fixed before the extrapolation point is determined in the extragradient framework, which prevents effective exploitation of local geometric structures and limits faster local convergence.

5. Numerical Experiments sec.experiments In this section, we present the numerical experiments to validate the global and local behavior of the proposed methods. All experiments are conducted on a single machine with a 14-core Apple M4 Pro CPU and 48GB LPDDR5 RAM. We conduct experiments on the regularized logistic regression problem, which is defined as follows:

$$f(x) = \frac{1}{N} \sum_{i=1}^N \log(1 + \exp(-b_i a_i^\top x)) + \frac{\gamma}{2} \|x\|^2, \quad \text{eq.logistic regression (5.1)}$$

where $a_i \in \mathbb{R}^n$ and $b_i \in \{-1, 1\}$, $\gamma = 10^{-4}$ is the regularization parameter. As mentioned, this problem is a notorious example where many accelerated SOMs are not as competitive as the classical Newton-type methods [7, 12, 70] numerically. Besides, a well-known estimate for the Lipschitz constant $\widehat{M}$ of $\nabla^2 f$ can be specified as follows:

$$\widehat{M} = \left\| \frac{1}{N} \sum_{i=1}^N a_i a_i^\top \right\| \max_{i \in [N]} \|a_i\|.$$

Although the estimate is conservative [59], we adopt it to isolate the basic algorithm frameworks from other practical enhancements, such as the adaptive adjustment of Lipschitz constants, see [9, 41].

We implement two different accelerated TR methods (Alg. 1 and Alg. 2) and compare them to some state-of-the-art SOMs, including

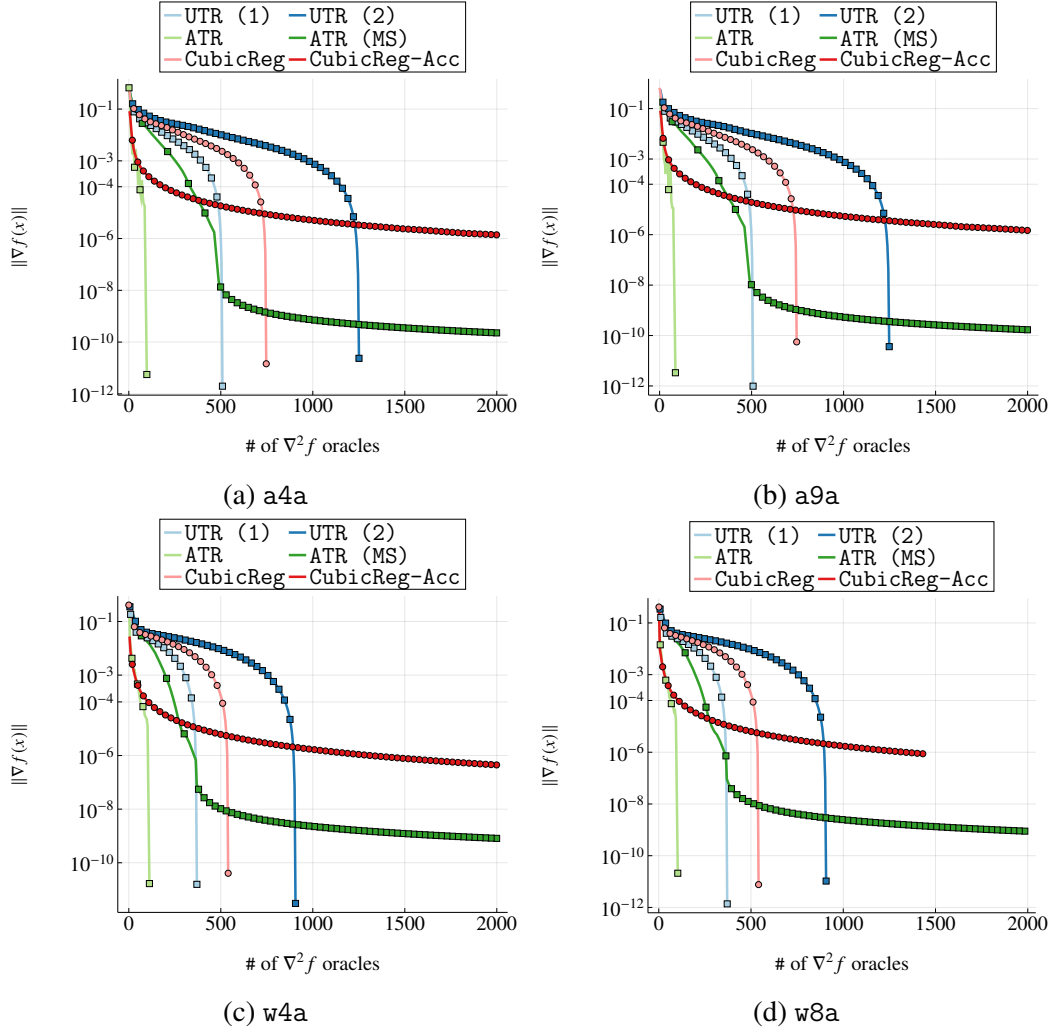


FIGURE 1. Logistic regression using the LIBSVM datasets `fig.logistic regression`

- The cubic regularized Newton method (CubicReg, Nesterov and Polyak [49]) and its accelerated version (CubicReg-A, Nesterov [46]).
- A non-accelerated TR method using the TR_+ oracle, by setting both (σ_k, r_k) proportionally to $\|\nabla f(x_k)\|^{1/2}$ similar to [35]. We test two non-accelerated TR methods UTR (1), UTR (2), using different Lipschitz estimates $\frac{\bar{M}}{2}, \bar{M}$, respectively. The purpose is to present the sensitivity of Lipschitz constants and set a fair comparison to CubicReg.

The subproblems arising in CubicReg and CubicReg-A are solved by a 1-D line-search strategy according to [47]. Similarly, TR_+ (in UTR, Alg. 1 and Alg. 2) is solved by searching the dual variable. All methods use exact Hessian evaluation and Cholesky factorization to solve the linear systems. Since these SOMs use different subproblems, and theoretically, the complexity rates to solve them vary from $O(\log(1/\epsilon))$ (for subproblems in a cubic regularized method [49]) to $O(\log \log(1/\epsilon))$ (for TR_+ [60, 65]), we report the number of Hessian evaluations needed in the method. In Figure 1, the performance of the SOMs on some LIBSVM datasets¹ is reported.

We could have several observations. Firstly, for non-accelerated methods, a trend of local superlinear convergence can be observed in UTR (1), UTR (2) and CubicReg. We could conclude that these three methods

¹ For details, see <https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/>

are comparable. Secondly, in the beginning of the iterations, all accelerated methods, including Alg. 1, Alg. 2, and CubicReg-A, converge faster than the non-accelerated methods (e.g., UTR (1), UTR (2), CubicReg). This confirms the effectiveness of the global acceleration.

Secondly, the global convergence does come at the expense of local superlinear convergence. Notably, Alg. 1 is the only accelerated method that has the local rate of superlinear convergence because of the diving track. Both Alg. 2 and CubicReg-A converge sublinearly in the local regime, in which Alg. 2 is slightly better because of its superior $\tilde{O}(\epsilon_f^{-2/7})$ non-asymptotic performance. These results are in accordance with what is predicted in the convergence analysis.

sec.proof pre

Appendix A: Technical proofs in Section 2

A.1. Auxiliary results of estimating sequence

lem.magnitude a and A

LEMMA A.1. For $k \geq 0$, we have

$$\frac{A_{k+1}}{a_k^{3/2}} \geq \frac{\sqrt{2}}{3}. \quad \text{eq.magnitude a and A (A.1)}$$

Proof. The conclusion follows from that

$$\frac{A_{k+1}}{a_k^{3/2}} = \frac{(k+1)(k+2)(k+3)}{6} \cdot \frac{2^{3/2}}{(k+1)^{3/2}(k+2)^{3/2}} = \frac{\sqrt{2}(k+3)}{3(k+1)^{1/2}(k+2)^{1/2}} \geq \frac{\sqrt{2}}{3}.$$

□

We call a differentiable function $d(x)$ on $\mathbb{R}^n$ uniformly convex [47, Section 4.2.2] of degree $p \geq 2$ with constant $q > 0$ if

$$d(y) \geq d(x) + \langle \nabla d(x), y - x \rangle + \frac{q}{p} \|y - x\|^p, \quad \forall x, y \in \mathbb{R}^n. \quad \text{eq.uniformly convex of p (A.2)}$$

lem.property of phi

LEMMA A.2. For the function sequence $\phi_k(x)$, they have the following properties

1. $\phi_k(x)$ is uniformly convex of degree 3 with constant $12M$.
2. v_k is the unique minimizer of $\phi_k(x)$, and

$$\phi_k(x) \geq \phi_k^* + 4M \|x - v_k\|^3, \quad \text{eq.phi_k phi_low (A.3)}$$

where $\phi_k^* = \min \phi_k(x)$.

Proof. The first claim is from the definition of ϕ_0 in (3.3), for analysis of uniform convex functions, please refer to [47, Section 4.2.2]. For the second claim, we have

$$\phi_k(x) = \phi_0(x) + \sum_{i=0}^{k-1} a_i (f(x_{i+1}) + \langle \nabla f(x_{i+1}), x - x_{i+1} \rangle). \quad \text{eq.phi recursive (A.4)}$$

From the optimality condition, we have that

$$\nabla \phi_k(x) = \nabla \phi_0(x) + \sum_{i=0}^{k-1} a_i \nabla f(x_{i+1}) = 24M \|x - x_0\| (x - x_0) + \sum_{i=0}^{k-1} a_i \nabla f(x_{i+1}) = 0.$$

Solving the optimality condition and noticing the way we update s_k gives

$$x = v_0 - \sqrt{\frac{1}{24M \|s_k\|}} s_k,$$

which is v_k . Then (A.3) holds from the fact that $\phi_k(x)$ is uniformly convex of degree 3. □

Proof to Lemma 2.2

Proof. From (2.5b), we have

$$\|\nabla f(x+d) - \nabla f(x) - \nabla^2 f(x)d\| \leq \frac{M}{2}\|d\|^2.$$

By (2.6) and the above, we have

$$\|\nabla f(x+d) + \mu d\| \leq \frac{M}{2}\|d\|^2. \quad \text{eq.bounded by squared step (A.5)}$$

Applying the triangle inequality and rearranging items, we can derive (2.7). $\square$

Proof to Lemma 2.3

Proof. Squaring both sides of (A.5) and rearranging items, we have

$$\begin{aligned} 2\mu \langle \nabla f(x+d), -d \rangle &\geq \|\nabla f(x+d)\|^2 + \mu^2\|d\|^2 - \frac{M^2}{4}\|d\|^4 \\ &\geq \|\nabla f(x+d)\|^2 + \frac{3}{4}\mu^2\|d\|^2. \end{aligned}$$

The second line is due to $\mu \geq M\|d\|$, dividing both sides by 2μ gives (2.8). Further, when $\mu \leq 2M\|d\|$, the above gives

$$4M\|d\| \langle \nabla f(x+d), -d \rangle \geq \|\nabla f(x+d)\|^2 + \frac{3}{4}M^2\|d\|^4,$$

which yields

$$\langle \nabla f(x+d), -d \rangle \geq \frac{\|\nabla f(x+d)\|^2}{4M\|d\|} + \frac{3}{16}M\|d\|^3.$$

Consider an auxiliary function $h(t) = \frac{\|\nabla f(x+d)\|^2}{4Mt} + \frac{3}{16}Mt^3$ where $t \geq 0$, taking derivatives gives $h(t)$ achieves its minimum at $t^* = \frac{\sqrt{2}\|\nabla f(x+d)\|^{1/2}}{\sqrt{3M}}$, plugging t^* back gives (2.9). $\square$

Appendix B: Technical proofs in Section 3

sec.proof variant 1

Proof to Lemma 3.6 To prove Lemma 3.6, we first introduce the following lemma, which discusses the property of the auxiliary function $g_y(\mu)$.

LEMMA B.1. *For any $y \in \mathbb{R}^n$, $g_y(\mu)$ is continuously differentiable and monotonically increasing for $\mu \in (0, +\infty]$ and*

$$g'_y(\mu) = \frac{1}{\|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\|} + \frac{\mu}{\|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\|^3} \sum_{i=1}^n \frac{\beta_i^2}{(\zeta_i + \mu)^3}, \quad \text{eq.derivative (B.1)}$$

and it is bounded both below and above

$$0 < g'_y(\mu) \leq \frac{2}{\|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\|}. \quad \text{eq.derivative bound (B.2)}$$

Where ζ_i is the i -th eigenvalue of $\nabla^2 f(y)$, $\beta_i = \nabla f(y)^T v_i$, v_i denotes the eigenvector corresponding to ζ_i .

Proof. Since $\nabla^2 f(y) + \mu I \succ 0$ for $\mu > 0$, we can apply eigen decomposition to it, then

$$\|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\| = \sqrt{\sum_{i=1}^n \frac{\beta_i^2}{(\zeta_i + \mu)^2}}, \quad g_y(\mu) = \frac{\mu}{\sqrt{\sum_{i=1}^n \frac{\beta_i^2}{(\zeta_i + \mu)^2}}}.$$

Obviously $g_y(\mu)$ is increasing in $\mu \in (0, +\infty]$. Also, through some basic calculations, we have

$$\begin{aligned} g'_y(\mu) &= \frac{1}{\sqrt{\sum_{i=1}^n \frac{\beta_i^2}{(\zeta_i + \mu)^2}}} + \mu \frac{1}{\left(\sum_{i=1}^n \frac{\beta_i^2}{(\zeta_i + \mu)^2}\right)^{3/2}} \sum_{i=1}^n \frac{\beta_i^2}{(\zeta_i + \mu)^3} \\ &= \frac{1}{\|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\|} + \frac{\mu}{\|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\|^3} \sum_{i=1}^n \frac{\beta_i^2}{(\zeta_i + \mu)^3} \\ &\leq \frac{1}{\|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\|} + \frac{\mu}{\|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\|^3} \sum_{i=1}^n \frac{\beta_i^2}{\mu (\zeta_i + \mu)^2} \\ &= \frac{2}{\|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\|}. \end{aligned}$$

The inequality is because of $\zeta_i \geq 0$ for $i = 1, \dots, n$. $\square$

Now we are ready to formally prove Lemma 3.6.

Proof. **seems requires $M > 1$** From the continuity and monotonicity of $g_y(\mu)$, we know there exists interval $[\mu_l, \mu_u]$ with (3.10) and $M \leq g_y(\mu) \leq 2M$ for all μ in this interval (intermediate value theorem).

Now we show that the length of the target interval for μ is bounded below; By the mean value theorem, there exists $\xi \in [\mu_l, \mu_u]$, such that

$$M = g_y(\mu_u) - g_y(\mu_l) = g'_y(\xi)(\mu_u - \mu_l),$$

combine the above with (B.2) and (3.10) we have

$$\mu_u - \mu_l = \frac{1}{g'_y(\xi)} (g_y(\mu_u) - g_y(\mu_l)) \geq \frac{M \|(\nabla^2 f(y) + \xi I)^{-1} \nabla f(y)\|}{2} \geq \frac{M \|(\nabla^2 f(y) + \mu_u I)^{-1} \nabla f(y)\|}{2}.$$

The third line is because of $\mu_u > \xi$. $\square$

Proof to Lemma 3.7

Proof. Denote the target interval for r as $[r_l, r_u]$ and

$$r_y(\mu) = \|(\nabla^2 f(y) + \mu I)^{-1} \nabla f(y)\|,$$

denote

$$r_l = r_y(\mu_u), \quad r_u = r_y(\mu_l).$$

The bisection terminates whenever $r \in [r_l, r_u]$. Note that by the mechanism of bisection, the oracle complexity is bounded by

$$O\left(\log \frac{r_+ - r_-}{r_u - r_l}\right) \leq O\left(\log \frac{r_+}{r_u - r_l}\right) \leq O\left(\log \frac{r_+}{r_y(\mu_l) - r_y(\mu_u)}\right).$$

For $r_y(\mu_l) - r_y(\mu_u)$, using mean value theorem, there exists $\xi \in [\mu_l, \mu_u]$ such that **check the second and third line, elaborate**

$$\begin{aligned} r_y(\mu_l) - r_y(\mu_u) &= r'_y(\xi)(\mu_l - \mu_u) \\ &= \text{Lemma B.1} \frac{\sum_{i=1}^n \frac{\beta_i^2}{(\zeta_i + \xi)^3}}{\|(\nabla^2 f(y) + \xi I)^{-1} \nabla f(y)\|} (\mu_u - \mu_l) \\ &\geq \frac{1}{\kappa_H + \mu_u} \|(\nabla^2 f(y) + \xi I)^{-1} \nabla f(y)\| (\mu_u - \mu_l) \\ &\geq \frac{1}{\kappa_H + \mu_u} \|(\nabla^2 f(y) + \mu_u I)^{-1} \nabla f(y)\|^2 \\ &\geq \frac{\|\nabla f(y)\|^2}{(\kappa_H + \mu_u)^3}. \end{aligned}$$

In the above equations, the second line is derived by taking derivative of $r_y(\mu)$, which is similar to the analysis in Lemma B.1. The third line is from Assumption 2.3. The fourth line is from (3.11). Therefore, we have

$$\begin{aligned}
 O\left(\log \frac{r_+}{r_y(\mu_l) - r_y(\mu_u)}\right) &\leq O\left(\log \frac{r_+(\kappa_H + \mu_u)^3}{\|\nabla f(y)\|^2}\right) \\
 &\leq O\left(\log \frac{\|\nabla f(y)\|(\kappa_H + \mu_u)^3}{M\|d_+\|\|\nabla f(y)\|^2}\right) \\
 &= O\left(\log \frac{(\kappa_H + \mu_u)^3}{M\|d_+\|\|\nabla f(y)\|}\right) \\
 &\leq_{ck} O\left(\log \frac{(\kappa_H + \mu_+)^3}{8M^2\|d_+\|^3}\right) \\
 &\leq_{ad} \max\left\{O\left(\log \frac{(\kappa_H + \mu_+)^3}{8\epsilon_g}\right), O\left(\log \frac{4\kappa_H^2(\kappa_H + \mu_+)^3}{8M\epsilon_g^2}\right)\right\} \\
 &= O\left(\log \frac{\kappa_H + \mu_+}{\epsilon_g}\right).
 \end{aligned}$$

In the above, the second line is from that the starting value as in Line 16 of Subroutine 1.

$$r_+ = \|(\nabla^2 f(y) + M\|d_+\|I)^{-1}\nabla f(y)\| \leq \frac{\|\nabla f(y)\|}{M\|d_+\|}.$$

The fourth line is from $\mu_u \leq \mu_+$ and from Line 9 in Alg. 1

$$\frac{\sqrt{2M}}{2}\|\nabla f(y)\|^{1/2} = \mu_+ \geq 2M\|d_+\|.$$

The last line is from the proposition of the logarithmic function. $\square$

Proof to Lemma 3.11

Proof. Let $R = \|x_0 - x^*\|$ and define

$$\rho := \min\left\{\frac{2\nu}{3M}, \frac{2\nu^2}{9\kappa_H M}\right\}.$$

It suffices to show $\|y_k - x^*\| < \rho$, since Assumption 2.3 and $\nabla f(x^*) = 0$ imply

$$\|\nabla f(y_k)\| = \|\nabla f(y_k) - \nabla f(x^*)\| \leq \kappa_H \|y_k - x^*\|.$$

We first derive two elementary bounds. From the estimating-sequence relation with $x = x^*$, we have, for every $k \geq 1$,

$$f(x_k) - f^* \leq \frac{\phi_0(x^*)}{A_k} = \frac{48MR^3}{k(k+1)(k+2)} \leq \frac{48MR^3}{k^3}.$$

Moreover, using (A.3) at $x = x^*$ and the estimating-sequence upper bound,

$$4M\|v_k - x^*\|^3 \leq \phi_k(x^*) - \phi_k^* \leq \phi_0(x^*) = 8MR^3.$$

Therefore,

$$\|v_k - x^*\| \leq 2^{1/3}R.$$

Next we show that x_k is already in $\mathcal{LQ}_p$ for the values of k considered below. For any x satisfying $\|x - x^*\| = 2\nu/(3M)$, Taylor's theorem and (3.17) give

$$f(x) - f^* \geq \frac{\nu}{6}\|x - x^*\|^2 = \frac{2\nu^3}{27M^2}.$$

By convexity, the same lower bound holds for every x with $\|x - x^*\| \geq 2\nu/(3M)$. Hence, if

$$f(x_k) - f^* < \frac{2\nu^3}{27M^2},$$

then $x_k \in \mathcal{LQ}_p$. In particular, the preceding global bound shows that this holds whenever

$$k \geq (648)^{1/3} \frac{MR}{\nu}.$$

For such k , (3.18) holds along the segment joining x_k and x^* , and hence

$$\frac{\nu}{6} \|x_k - x^*\|^2 \leq f(x_k) - f^* \leq \frac{48MR^3}{k^3}.$$

Thus

$$\|x_k - x^*\| \leq \sqrt{\frac{288MR^3}{\nu k^3}}.$$

Using the definition of y_k ,

$$y_k = \frac{k}{k+3} x_k + \frac{3}{k+3} \nu_k,$$

we obtain

$$\begin{aligned} \|y_k - x^*\| &\leq \frac{k}{k+3} \|x_k - x^*\| + \frac{3}{k+3} \|\nu_k - x^*\| \\ &\leq \sqrt{\frac{288MR^3}{\nu k^3}} + \frac{3 \cdot 2^{1/3} R}{k}. \end{aligned}$$

Finally, since Assumption 2.3 and Assumption 3.1 imply $\kappa_H \geq \nu$, we have

$$\rho = \min \left\{ \frac{2\nu}{3M}, \frac{2\nu^2}{9\kappa_H M} \right\} = \frac{2\nu^2}{9\kappa_H M}.$$

If

$$k \geq 36 \frac{\kappa_H MR}{\nu^2},$$

then this condition also implies

$$k \geq (648)^{1/3} \frac{MR}{\nu},$$

and the two terms above satisfy

$$\sqrt{\frac{288MR^3}{\nu k^3}} \leq \frac{\rho}{2}, \quad \frac{3 \cdot 2^{1/3} R}{k} \leq \frac{\rho}{2}.$$

Consequently,

$$\|y_k - x^*\| \leq \rho.$$

Therefore,

$$\|y_k - x^*\| \leq \frac{2\nu}{3M}, \quad \|\nabla f(y_k)\| \leq \kappa_H \|y_k - x^*\| \leq \frac{2\nu^2}{9M},$$

which means $y_k \in \mathcal{LQ}_g$. $\square$

Lemma B.2 and its proof

LEMMA B.2. *Let $\{a_i\}$ satisfy $a_i \leq C a_{i-1}^2$ with $C > 0$ and $0 < Ca_0 < 1$. Let $\eta_0 = Ca_0$ and define*

$$\epsilon^* = \frac{1}{C} \left(\eta_0 \ln \frac{1}{\eta_0} \right)^2.$$

Let $i_{\min} = \min\{i \in \mathbb{N} : a_i \leq \epsilon\}$. Then for all $0 < \epsilon \leq \epsilon^$,*

$$i_{\min} \leq \left\lceil \log_2 \frac{a_0}{\epsilon} \right\rceil.$$

Proof. From the quadratic recurrence, we have the bound

$$a_i \leq C^{2^i-1} a_0^{2^i} = \frac{(Ca_0)^{2^i}}{C} = \frac{\eta_0^{2^i}}{C}.$$

Thus, $a_i \leq \epsilon$ is guaranteed if $\eta_0^{2^i} \leq C\epsilon$, which is equivalent to

$$2^i \geq \frac{\ln(1/(C\epsilon))}{\ln(1/\eta_0)}.$$

It follows that

$$i_{\min} \leq \left\lceil \log_2 \left(\frac{\ln(1/(C\epsilon))}{\ln(1/\eta_0)} \right) \right\rceil.$$

To bound this by $\lceil \log_2(a_0/\epsilon) \rceil$, it suffices to show

$$\frac{\ln(1/(C\epsilon))}{\ln(1/\eta_0)} \leq \frac{a_0}{\epsilon} \iff w \ln \frac{1}{w} \leq \eta_0 \ln \frac{1}{\eta_0},$$

where $w := C\epsilon$. Let $K := \eta_0 \ln(1/\eta_0)$. Since $\eta_0 \in (0, 1)$, $K \leq 1/e$.

For any $\epsilon \leq \epsilon^*$, we have $w = C\epsilon \leq K^2 \leq (1/e)^2 < 1/e$. Because the function $x \mapsto x \ln(1/x)$ is strictly increasing on $(0, 1/e]$, evaluating it at K^2 yields

$$w \ln \frac{1}{w} \leq K^2 \ln \left(\frac{1}{K^2} \right) = 2K \left(K \ln \frac{1}{K} \right).$$

By the maximum property of the function $x \ln(1/x)$, we have $K \ln(1/K) \leq 1/e$. Substituting this gives

$$w \ln \frac{1}{w} \leq \frac{2}{e} K < K = \eta_0 \ln \frac{1}{\eta_0}.$$

This confirms the desired inequality. $\square$

sec.appendix 2

Appendix C: Technical proofs in Section 4

sec.app-c

Proof to Lemma 4.3

Proof. First note Line 9 in Alg. 2. For $i \geq 0$

$$A_{i+1}^{1/2} - A_i^{1/2} = \frac{a_i}{A_{i+1}^{1/2} + A_i^{1/2}} = \frac{1}{A_{i+1}^{1/2} + A_i^{1/2}} \sqrt{\frac{A_{i+1}}{\sigma_i}} \geq \frac{1}{2\sqrt{\sigma_i}}.$$

The second equality comes from $\sigma_i a_i^2 = A_i + a_i = A_{i+1}$ ((4.2)). Summing up the above from $i = 0$ to $k - 1$ gives

$$A_k \geq \frac{1}{4} \left(\sum_{i=0}^{k-1} \frac{1}{\sigma_i^{1/2}} \right)^2,$$

from (4.3), the above gives

$$A_k \geq \frac{\eta}{4M} \left(\sum_{i=0}^{k-1} \frac{1}{\|d_i\|^{1/2}} \right)^2, \quad \text{eq. Ak helper 1 (C.1)}$$

on the other hand, from (4.5) we have

$$B_k = \frac{3\gamma M}{8} \sum_{i=0}^{k-1} A_{i+1} \|d_i\|^3 \leq \frac{1}{2} \|v_0 - x^*\|^2.$$

To estimate A_k from below, define $\zeta_i = \|d_i\|^{1/2}$, $D = \frac{4}{3\gamma M} \|v_0 - x^*\|^2$, we use the following auxiliary optimization problem

$$\zeta^* = \min_{\zeta \in \mathbb{R}^k} \left\{ \sum_{i=0}^{k-1} \frac{1}{\zeta_i} : \sum_{i=0}^{k-1} A_{i+1} \zeta_i^6 \leq D \right\}.$$

Introducing the Lagrangian multiplier w , the optimality condition gives

$$\frac{1}{\zeta_i^2} = 6w A_{i+1} \zeta_i^5, \quad i = 0, \dots, k-1$$

thus $\zeta_i = \left(\frac{1}{wA_{i+1}}\right)^{1/7}$, we have $w > 0$ and the constraint is active,

$$D = \sum_{i=0}^{k-1} A_{i+1} \left(\frac{1}{wA_{i+1}}\right)^{6/7} = \frac{1}{w^{6/7}} \sum_{i=0}^{k-1} A_{i+1}^{1/7},$$

therefore $\zeta^* = \sum_{i=0}^{k-1} (wA_{i+1})^{1/7} = \frac{1}{D^{1/6}} \left(\sum_{i=0}^{k-1} A_{i+1}^{1/7}\right)^{7/6}$, plugging back $\|d_i\|$ and $\|v_0 - x^*\|^2$, we have

$$A_k \geq \sum_{i=0}^{k-1} \frac{1}{\|d_i\|^{1/2}} \geq \left(\frac{3\gamma M}{4\|v_0 - x^*\|^2}\right)^{1/6} \left(\sum_{i=0}^{k-1} A_{i+1}^{1/7}\right)^{7/6},$$

from (C.1) we have

$$A_k \geq \frac{\eta}{4M} \left(\frac{3\gamma M}{4\|v_0 - x^*\|^2}\right)^{1/3} \left(\sum_{i=1}^k A_i^{1/7}\right)^{7/3}, \quad k \geq 1. \quad \text{eq. Ak helper 2 (C.2)}$$

Denote $\omega = \frac{\eta}{4M} \left(\frac{3\gamma M}{4\|v_0 - x^*\|^2}\right)^{1/3}$, $C_k = \left(\sum_{i=1}^k A_i^{1/7}\right)^{2/3}$, plugging them into (C.2) we have

$$C_1 \geq \omega^{1/7}, \quad C_{k+1}^{3/2} - C_k^{3/2} \geq \omega^{1/7} C_{k+1}^{1/2},$$

which gives

$$\begin{aligned} \omega^{1/7} C_{k+1}^{1/2} &\leq \left(C_{k+1}^{1/2} - C_k^{1/2}\right) \left(C_{k+1}^{1/2} \left(C_{k+1}^{1/2} + C_k^{1/2}\right) + C_k\right) \\ &\leq \left(C_{k+1}^{1/2} - C_k^{1/2}\right) \left(C_{k+1}^{1/2} \left(C_{k+1}^{1/2} + C_k^{1/2}\right) + \frac{1}{2} C_{k+1}^{1/2} \left(C_{k+1}^{1/2} + C_k^{1/2}\right)\right) \\ &= \frac{3}{2} C_{k+1}^{1/2} (C_{k+1} - C_k). \end{aligned}$$

Thus $C_k \geq \omega^{1/7} \left(1 + \frac{2}{3}(k-1)\right)$, $k \geq 1$. For A_k , by (C.2) we have

$$\begin{aligned} A_k &\geq \omega \left(C_k^{3/2}\right)^{7/3} \geq \omega \left(\omega^{1/7} \cdot \frac{2k+1}{3}\right)^{7/2} = \omega^{3/2} \left(\frac{2k+1}{3}\right)^{7/2} \\ &= \left(\frac{\eta}{4} \left(\frac{3\gamma}{4M^2\|v_0 - x^*\|^2}\right)^{1/3}\right)^{3/2} \left(\frac{2k+1}{3}\right)^{3.5} \end{aligned}$$

□

Properties of $\psi(\sigma, y)$ Now we introduce some basic properties of $\psi(\sigma, y)$:

LEMMA C.1. For any $y \in \mathbb{R}^n$, suppose $0 < \sigma \leq \bar{\sigma}$, then we have

$$\left(\frac{\sigma}{\bar{\sigma}}\right)^2 \psi(\sigma, y) \leq \psi(\bar{\sigma}, y) \leq \frac{\sigma}{\bar{\sigma}} \psi(\sigma, y). \quad \text{eq. psi property wrt sigma (C.3)}$$

Proof. Note that

$$\sigma \psi(\sigma, y) = \left\| \left(\nabla^2 f(y) + \sigma I \right)^{-1} \nabla f(y) \right\|.$$

Since $\nabla^2 f(y) + \bar{\sigma} I \succeq \nabla^2 f(y) + \sigma I \succ 0$, we have

$$\sigma \psi(\sigma, y) \geq \bar{\sigma} \psi(\bar{\sigma}, y),$$

which is the second argument. Similarly,

$$\sigma^2 \psi(\sigma, y) = \left\| \left(\frac{\nabla^2 f(y)}{\sigma} + I \right)^{-1} \nabla f(y) \right\|,$$

since $\frac{1}{\sigma} \nabla^2 f(y) + I \succeq \frac{1}{\bar{\sigma}} \nabla^2 f(y) + I \succ I$, we have $\sigma^2 \psi(\sigma, y) \leq \bar{\sigma}^2 \psi(\bar{\sigma}, y)$, which finished the proof. □

LEMMA C.2. Suppose Assumption 2.1 holds, for any $y, \bar{y} \in \mathbb{R}^n$ and $\sigma > 0$, then ^{lem.psi property wrt y}

$$|\psi(\sigma, y) - \psi(\sigma, \bar{y})| \leq \frac{1}{\sigma} \|y - \bar{y}\| + \frac{M}{\sigma^2} \|y - \bar{y}\|^2 + \frac{2M}{\sigma} \|y - \bar{y}\| \delta, \quad \text{eq.psi property wrt y (C.4)}$$

where $\delta := \min\{\psi(\sigma, \bar{y}), \psi(\sigma, y)\}$. Further, we have

$$\psi(\sigma, y) \leq \frac{1}{\sigma} \|y - \bar{y}\| + \frac{M}{\sigma^2} \|y - \bar{y}\|^2 + \left(\frac{2M}{\sigma} \|y - \bar{y}\| + 1 \right) \psi(\sigma, \bar{y}). \quad \text{eq.psi property wrt y advanced (C.5)}$$

Proof. We denote

$$\begin{aligned} x &= \arg \min_{x \in \mathbb{R}^n} \nabla f(y)^T(x - y) + \frac{1}{2}(x - y)\nabla^2 f(y)(x - y) + \frac{\sigma}{2} \|x - y\|^2, \\ \bar{x} &= \arg \min_{x \in \mathbb{R}^n} \nabla f(\bar{y})^T(x - \bar{y}) + \frac{1}{2}(x - \bar{y})\nabla^2 f(\bar{y})(x - \bar{y}) + \frac{\sigma}{2} \|x - \bar{y}\|^2, \end{aligned}$$

the optimality conditions of the above are

$$\nabla f(y) + \nabla^2 f(y)(x - y) + \sigma(x - y) = 0, \quad \nabla f(\bar{y}) + \nabla^2 f(\bar{y})(\bar{x} - \bar{y}) + \sigma(\bar{x} - \bar{y}) = 0.$$

Denote

$$v = \nabla f(y) + \nabla^2 f(y)(x - y), \quad \bar{v} = \nabla f(\bar{y}) + \nabla^2 f(\bar{y})(\bar{x} - \bar{y}), \quad u = \nabla f(\bar{y}) + \nabla^2 f(\bar{y})(x - \bar{y}).$$

Plugging them into the optimality conditions gives

clarify C8

$$\begin{aligned} v + \sigma(x - y) &= 0, & \text{eq.def } v \text{ (C.6)} \\ \bar{v} + \sigma(\bar{x} - \bar{y}) &= 0, & \text{eq.def } \bar{v} \text{ (C.7)} \end{aligned}$$

$$|\psi(\sigma, y) - \psi(\sigma, \bar{y})| = \left| \frac{1}{\sigma^2} \|v\| - \frac{1}{\sigma^2} \|\bar{v}\| \right| \leq \frac{1}{\sigma^2} \|v - \bar{v}\|. \quad \text{eq.target distance initial (C.8)}$$

Subtracting (C.7) from (C.6), we have

$$\sigma(x - \bar{x}) + v - \bar{v} = \sigma(y - \bar{y}).$$

Plus both sides by u , we have

$$\sigma(x - \bar{x}) + u - \bar{v} = \sigma(y - \bar{y}) + u - v.$$

It is easy to show that

$$\langle x - \bar{x}, u - \bar{v} \rangle = \langle x - \bar{x}, \nabla^2 f(\bar{y})(x - \bar{x}) \rangle \geq 0,$$

thus from triangle inequality, we have

$$\|u - \bar{v}\| \leq \|\sigma(x - \bar{x}) + u - \bar{v}\| \leq \sigma \|y - \bar{y}\| + \|u - v\|,$$

hence

$$\|v - \bar{v}\| \leq \sigma \|y - \bar{y}\| + 2\|u - v\|. \quad \text{eq.bound psi 1 (C.9)}$$

Now we bound $\|u - v\|$,

$$\begin{aligned} \|u - v\| &= \|\nabla f(\bar{y}) + \nabla^2 f(\bar{y})(x - \bar{y}) - (\nabla f(y) + \nabla^2 f(y)(x - y))\| \\ &= \|(\nabla f(\bar{y}) + \nabla^2 f(\bar{y})(y - \bar{y}) - \nabla f(y)) + (\nabla^2 f(\bar{y})(x - y) - \nabla^2 f(y)(x - y))\| \\ &\leq \|(\nabla f(\bar{y}) + \nabla^2 f(\bar{y})(y - \bar{y}) - \nabla f(y))\| + \|(\nabla^2 f(\bar{y}) - \nabla^2 f(y))(x - y)\| \\ &\leq \frac{M}{2} \|y - \bar{y}\|^2 + M \|y - \bar{y}\| \|x - y\| = \frac{M}{2} \|y - \bar{y}\|^2 + \sigma M \|y - \bar{y}\| \psi(\sigma, y). \end{aligned} \quad \text{eq.bound psi 2 (C.10)}$$

Plugging (C.9) and (C.10) into (C.8), we have

$$|\psi(\sigma, y) - \psi(\sigma, \bar{y})| \leq \frac{1}{\sigma} \|y - \bar{y}\| + \frac{M}{\sigma^2} \|y - \bar{y}\|^2 + \frac{2M}{\sigma} \|y - \bar{y}\| \psi(\sigma, y).$$

Similarly, we can prove

$$|\psi(\sigma, y) - \psi(\sigma, \bar{y})| \leq \frac{1}{\sigma} \|y - \bar{y}\| + \frac{M}{\sigma^2} \|y - \bar{y}\|^2 + \frac{2M}{\sigma} \|y - \bar{y}\| \psi(\sigma, \bar{y}).$$

Hence we have proved (C.4), by applying triangle inequality, we have (C.5). $\square$

Proof to Lemma 4.4

Proof. If (4.8) holds, we have

$$\|d\| = \|(\nabla^2 f(y(\sigma)) + \sigma I)^{-1} \nabla f(y(\sigma))\| < \frac{\eta}{M} \sigma,$$

which is $\psi(\sigma, y(\sigma)) < \frac{\eta}{M}$. Else if (4.9) holds, we have

$$\|d\| = \|(\nabla^2 f(y(\sigma)) + \sigma I + \lambda I)^{-1} \nabla f(y(\sigma))\| = \frac{1}{M} \sigma.$$

By (C.3), we have

$$\|(\nabla^2 f(y(\sigma)) + \sigma I)^{-1} \nabla f(y(\sigma))\| > \|(\nabla^2 f(y(\sigma)) + \sigma I + \lambda I)^{-1} \nabla f(y(\sigma))\| = \frac{1}{M} \sigma,$$

which means $\psi(\sigma, y(\sigma)) > \frac{1}{M}$. $\square$

Proof to Lemma 4.6 In fact, we prove a more detailed version of Lemma 4.6 here, which is lemma C.5 here. First, we introduce the following property of $y(\cdot)$, since it is an auxiliary lemma, and the proof is almost the same as in Monteiro and Svaiter [42].

LEMMA C.3 (Lemma 7.13, Monteiro and Svaiter [42]). *Suppose Assumption 2.2 holds, and there exists $M_0 > 0$ such that*

$$\|x - x^*\| \leq M_0, \|v - x^*\| \leq M_0, \quad \text{lem.curve condition} \quad \text{eq.bounded assm point} \quad (\text{C.11})$$

then the curve $y(\cdot)$ satisfies

$$\|y(s) - y(t)\| \leq \frac{M_0}{t} (s - t), \quad \forall s \geq t > 0. \quad \text{eq.curve condition} \quad (\text{C.12})$$

Proof. We have

$$y(\sigma) = \frac{A}{A + a(\sigma)} x + \frac{a(\sigma)}{A + a(\sigma)} v = x + \tau(\sigma)(v - x),$$

where $\tau(\sigma) = \frac{a(\sigma)}{A + a(\sigma)}$. For any $s \geq t > 0$,

$$\|y(s) - y(t)\| = |\tau(s) - \tau(t)| \|v - x\|,$$

by the mean value theorem, we have

$$\|y(s) - y(t)\| = |\tau'(\xi)| (s - t) \|v - x\| \leq M_0 |\tau'(\xi)| (s - t),$$

where $\xi \in [t, s]$. Note that from (4.2) we have $\sigma a^2 = a + A$, it leads to

$$\tau(\sigma) = \frac{a(\sigma)}{A + a(\sigma)} = \frac{1}{\sigma a(\sigma)} = \frac{2}{1 + \sqrt{1 + 4A\sigma}}.$$

also review the constant term in the statement

Its derivative is

$$\tau'(\sigma) = -\frac{4A}{\sqrt{1 + 4A\sigma} (1 + \sqrt{1 + 4A\sigma})^2},$$

therefore for all $\sigma > 0$, we have

$$|\tau'(\sigma)| \leq \frac{2A}{(1 + \sqrt{1 + 4A\sigma})^2} \leq \frac{2A}{4A\sigma} = \frac{1}{2\sigma}.$$

Therefore we have

$$\begin{aligned}\|y(s) - y(t)\| &\leq |\tau'(\xi)| (s - t) \|x - v\| \\ &\leq |\tau'(\xi)| (s - t) (\|x - x^*\| + \|v - x^*\|) \\ &\leq 2M_0 |\tau'(\xi)| (s - t) \\ &\leq \frac{M_0}{t} (s - t).\end{aligned}$$

□

Next, we proceed to introduce the lemma with analyzes the difference between ψ_- and ψ_+ :

LEMMA C.4. *Suppose Assumption 2.1 and Assumption 2.2 hold, and there exists $M_0 > 0$ such that (C.11) holds. Then in R&B (Subroutine 2), we have*

$$\begin{aligned}\psi_- - \psi_+ &\leq \left(\frac{\sigma_+}{\sigma_-}\right)^2 \left[\frac{1}{\sigma_+} \|y(\sigma_-) - y(\sigma_+)\| + \frac{M}{\sigma_+^2} \|y(\sigma_-) - y(\sigma_+)\|^2 \right. \\ &\quad \left. + \frac{2M}{\sigma_+} \|y(\sigma_-) - y(\sigma_+)\| \cdot \psi_+ \right] + \left[\left(\frac{\sigma_+}{\sigma_-}\right)^2 - 1 \right] \psi_+.\end{aligned}\tag{C.13}$$

Further, it gives

$$\psi_- - \psi_+ \leq \frac{\sigma_+}{\sigma_-^2} \left[\frac{M_0}{\sigma_-} + \frac{MM_0^2}{\sigma_-^2} + 2 \left(\frac{MM_0}{\sigma_-} + 1 \right) \psi_+ \right] + \left[\left(\frac{\sigma_+}{\sigma_-}\right)^2 - 1 \right] \psi_+.\tag{C.14}$$

Proof.

$$\begin{aligned}\psi_- - \psi_+ &= \psi(\sigma_-, y(\sigma_-)) - \psi(\sigma_+, y(\sigma_+)) \\ &\leq \left(\frac{\sigma_+}{\sigma_-}\right)^2 \psi(\sigma_+, y(\sigma_-)) - \psi(\sigma_+, y(\sigma_+)) \\ &\leq \left(\frac{\sigma_+}{\sigma_-}\right)^2 \left[\frac{1}{\sigma_+} \|y(\sigma_-) - y(\sigma_+)\| + \frac{M}{\sigma_+^2} \|y(\sigma_-) - y(\sigma_+)\|^2 + \left(\frac{2M}{\sigma_+} \|y(\sigma_-) - y(\sigma_+)\| + 1 \right) \psi_+ \right] - \psi_+ \\ &= \left(\frac{\sigma_+}{\sigma_-}\right)^2 \left[\frac{1}{\sigma_+} \|y(\sigma_-) - y(\sigma_+)\| + \frac{M}{\sigma_+^2} \|y(\sigma_-) - y(\sigma_+)\|^2 + \frac{2M}{\sigma_+} \|y(\sigma_-) - y(\sigma_+)\| \psi_+ \right] \\ &\quad + \left[\left(\frac{\sigma_+}{\sigma_-}\right)^2 - 1 \right] \psi_+.\end{aligned}$$

The second line is because of Lemma C.1, the third line is because of Lemma C.2. To derive (C.14), just plug (C.12) into (C.13). □

Now we can proceed with the main proof.

LEMMA C.5. *recompute the bound*

Suppose Assumption 2.1 and Assumption 2.2 hold, and there exists $M_0 > 0$ such that (C.11) holds. Then in R&B, we have

$$\begin{aligned}&\frac{\sigma_+}{\sigma_-^2} \left[\frac{M_0}{\sigma_-} + \frac{MM_0^2}{\sigma_-^2} + 2 \left(\frac{MM_0}{\sigma_-} + 1 \right) \psi_+ \right] \\ &\leq \sqrt{G_0} \left(\frac{(1+2\theta)^{3/2} M_0}{2\sqrt{2}M\eta\epsilon_g^{3/2}} + \frac{(1+2\theta)^2 M_0^2}{4\sqrt{M}\eta\epsilon_g^2} + \frac{(1+2\theta)^{3/2} M_0}{\sqrt{2}M\epsilon_g^{3/2}} + \frac{(1+2\theta)\sqrt{G_0}}{M^{3/2}\epsilon_g} \right).\end{aligned}$$

Denote

$$C(G_0, M_0, M, \epsilon_g) := M\sqrt{G_0} \left(\frac{(1+2\theta)^{3/2} M_0}{2\sqrt{2}M\eta\epsilon_g^{3/2}} + \frac{(1+2\theta)^2 M_0^2}{4\sqrt{M}\eta\epsilon_g^2} + \frac{(1+2\theta)^{3/2} M_0}{\sqrt{2}M\epsilon_g^{3/2}} + \frac{(1+2\theta)\sqrt{G_0}}{M^{3/2}\epsilon_g} \right),$$

we have

$$\sigma_+ - \sigma_- > \frac{1 - \eta}{C(G_0, M_0, M, \epsilon_g)}. \quad \text{eq. bound value by brackets (C.15)}$$

Further, the oracle called during the bisection is bounded by

$$\log \left(\frac{\sqrt{\frac{MG_0}{\eta}} C(G_0, M_0, M, \epsilon_g)}{1 - \eta} \right).$$

Proof. Note that if the bisection procedure does not terminate, from (C.15) we have

$$\sigma_+ - \sigma_- > \frac{1 - \eta}{C(G_0, M_0, M, \epsilon_g)}.$$

Note that the bracket points are defined as in (4.12), therefore by the mechanism of the bisection method, the total number of bisection in the k -th iteration N_k is bounded by

$$N_k \leq \log \left(\frac{\left(\sqrt{\frac{MG_0}{\eta}} - \sqrt{\frac{2M\epsilon_g}{1+2\theta}} \right) C(G_0, M_0, M, \epsilon_g)}{1 - \eta} \right) \leq \log \left(\frac{\sqrt{\frac{MG_0}{\eta}} C(G_0, M_0, M, \epsilon_g)}{1 - \eta} \right).$$

Since G_0, M_0 is polynomial in D_0 , by the definition of $C(G_0, M_0, M, \epsilon_g)$ and omitting the algorithm parameters, we conclude $N_k \leq O(\log MD_0/\epsilon_g)$. $\square$

Proof to Lemma 4.7

Proof. First, we show that (4.18) is a direct consequence of (4.17):

$$\|\nabla f(y_k(\sigma)) - \nabla f(x^*) - \nabla^2 f(x^*)(y_k(\sigma) - x^*)\| \leq \frac{M}{2} \|y_k(\sigma) - x^*\|^2.$$

Using (2.5a), by the triangle inequality, we have

$$\begin{aligned} \|\nabla f(y_k(\sigma))\| &\leq \|\nabla f(x^*) + \nabla^2 f(x^*)(y_k(\sigma) - x^*)\| + \frac{M}{2} \|y_k(\sigma) - x^*\|^2 \\ &\leq \|\nabla^2 f(x^*)(y_k(\sigma) - x^*)\| + \frac{M}{2} \|y_k(\sigma) - x^*\|^2 \\ &\leq \left(\frac{4}{\sqrt{3\gamma}} + 1 \right) \|\nabla^2 f^*\| D_0 + \frac{M}{2} \left(\frac{4}{\sqrt{3\gamma}} + 1 \right)^2 D_0^2. \end{aligned}$$

Therefore, we only need to prove (4.16) and (4.17) hold. We prove them by induction. It is trivial that they hold for $i = 0$. Suppose that they hold for $i = k$. We will prove that they also hold for $i = k + 1$. Note that

$$\begin{aligned} \|x_{k+1} - x^*\| &\leq \|x_{k+1} - y_k\| + \|y_k - x^*\| \\ &\leq \|d_k\| + \frac{A_k}{A_{k+1}} \|x_k - x^*\| + \frac{a_k}{A_{k+1}} \|v_k - x^*\| \\ &\leq \frac{1}{A_{k+1}} (A_{k+1} \|d_k\| + A_k \|x_k - x^*\| + a_k D_0) \\ &\leq \frac{1}{A_{k+1}} \left(\sum_{i=0}^k A_{i+1} \|d_i\| \right) + D_0. \end{aligned}$$

The second line is from (4.1), the third line is from (4.5), and the fourth line is derived by iterating

$$A_{k+1} \|x_{k+1} - x^*\| \leq A_{k+1} \|d_k\| + A_k \|x_k - x^*\| + a_k D_0,$$

which is derived by multiplying both sides of the third line by A_{k+1} .

Summing up (4.4), we have

$$\frac{3\gamma}{8} \sum_{i=0}^k A_{i+1} \sigma_i \|d_i\|^2 \leq \frac{1}{2} D_0^2,$$

to bound $\sum_{i=0}^k A_{i+1} \|d_i\|$, we come to the optimization problem:

$$\max_{\zeta \in \mathbb{R}_+^{k+1}} \left\{ \sum_{i=0}^k A_{i+1} \zeta_i : \sum_{i=0}^k A_{i+1} \sigma_i \zeta_i^2 \leq \frac{4}{3\gamma} D_0^2 \right\},$$

through similar analysis in Monteiro and Svaiter [42, Lemma A.2] we have

$$\sum_{i=0}^k A_{i+1} \|d_i\| \leq \sqrt{\frac{4}{3\gamma}} \cdot \sqrt{\sum_{i=0}^k \frac{A_{i+1}}{\sigma_i}} D_0,$$

therefore,

$$\|x_{k+1} - x^*\| \leq \frac{1}{A_{k+1}} \left(\sum_{i=0}^k A_{i+1} \|d_i\| \right) + D_0 \leq \left(\sqrt{\frac{4}{3\gamma}} \cdot \sum_{i=0}^k \sqrt{\frac{1}{\sigma_i}} \cdot \sqrt{\frac{1}{A_{k+1}}} + 1 \right) D_0. \quad \text{eq} \quad \text{bound initial } x \quad (\text{C.16})$$

The second inequality is from the fact that A_k is monotone and 2-norm is majorized by 1-norm. From (4.2) we have

$$a_k \geq \frac{1}{2\sigma_k} + \sqrt{\frac{A_k}{\sigma_k}},$$

hence

$$A_{k+1} \geq A_k + \frac{1}{2\sigma_k} + \sqrt{\frac{A_k}{\sigma_k}} \geq A_k + \frac{1}{4\sigma_k} + \sqrt{\frac{A_k}{\sigma_k}}.$$

Taking the square root of both sides,

$$\sqrt{A_{k+1}} \geq \sqrt{A_k} + \frac{1}{2\sqrt{\sigma_k}},$$

Iterating the above inequality gives

$$\sqrt{A_{k+1}} \geq \sum_{i=0}^k \frac{1}{2\sqrt{\sigma_i}}.$$

Plugging the above into (C.16), we have

$$\|x_{k+1} - x^*\| \leq \left(\frac{4}{\sqrt{3\gamma}} + 1 \right) D_0.$$

For v_{k+1} , we have (4.16), (4.17) and (4.18) hold for iteration k . As a result, the bisection search procedure is valid and (4.3) holds for the k -th iteration due to Corollary C.5. Hence $\|v_{k+1} - x^*\| \leq D_0$.

To prove the boundedness of $y_{k+1}(\sigma)$, just note that $y_{k+1}(\sigma)$ is a linear combination of v_{k+1} and x_{k+1} .
□

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